

False message detection in Internet of Vehicle through machine learning and vehicle consensus

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ABSTRACT

The introduction of Internet of Vehicles (IoV) is promising in enabling vehicles to communicate with each other. However, attackers can invade the on-board operating systems and send fabricated messages, compromising the reliability of IoV. Controlled by attackers, malicious vehicles can send falsified data in Basic Safety Messages (BSM), which are periodically-sent beacon messages for traffic management. Also, Event Report Messages (ERM) can be forged to mislead other vehicles about the status of a traffic event. To alleviate false message injections in the IoV, this paper proposes a detection scheme by exploring traffic data and vehicle consensus. On the one hand, our scheme leverages the fact that traffic events may have an influence on vehicle movements. In detail, the vehicle densities in an area are processed as time series data. Then, an anomaly detection model based on the Gradient Boosting Decision Tree (GBDT) is designed to calculate the traffic anomaly index. In situations where the traffic anomaly is insignificant, a vehicle consensus mechanism is further integrated into the detection scheme. On the other hand, data clustering techniques are applied to separate malicious and benign vehicles in terms of false BSMs. The distance of each vehicle to its cluster centroid is combined with hypothesis testing to measure the cluster quality. The effectiveness of our proposed scheme is validated by extensive simulations based on a realistic road network. Our proposed scheme outperforms several baselines for both false BSM and ERM detections in terms of precision, recall, and F1 score.

1. Introduction

The advent of Internet of Things (IoT) has revolutionized various domains, and one of the most promising applications is the Internet of Vehicles (IoV). The IoV encompasses the integration of vehicles, infrastructure, and other road users into a network, enabling seamless communication and exchange of information. Specifically, the IoV mainly consists of five types of communications (Ji et al., 2020): Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), Vehicle-to-Road (V2R), Vehicle-to-Sensor (V2S) and Vehicle-to-Pedestrian (V2P). This interconnectedness has paved the way for numerous advancements, including intelligent transportation systems, non-line-of-sight perception, and autonomous driving (Duan et al., 2020). Combined with other enabling technologies such as edge computing (Zhang, Zhang et al., 2024), the IoV also enables efficient traffic management (Sheikh &

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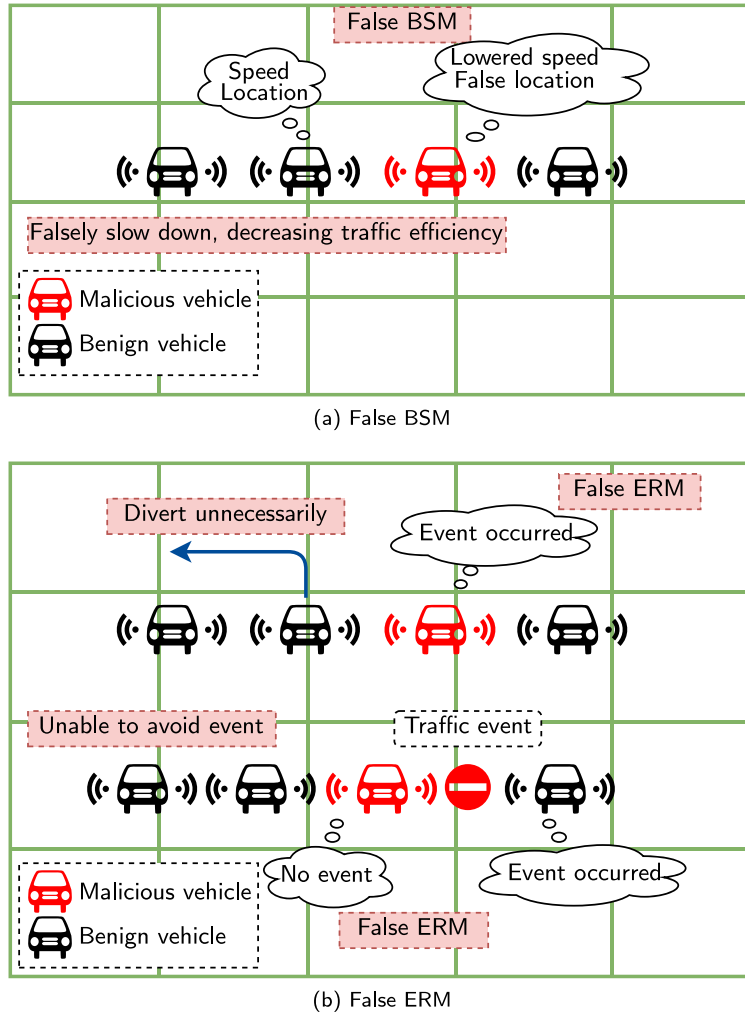


Fig. 1. The illustration of two types of false messages.

Liang, 2019), in which traffic information is shared through the IoV between vehicles and other network participants. For instance, by sharing road traffic conditions in real time, when a traffic accident occurs ahead, the vehicle in front will promptly transmit the information to the vehicle behind to avoid secondary accidents (Ji et al., 2020). Moreover, in emergency situations such as earthquakes, the IoV can be utilized to disseminate alerts and environmental information through multi-hop communications (Grassi et al., 2023).

However, as the IoV continues to evolve, security issues have emerged as a paramount concern for researchers and practitioners. The integration of vehicles into a network exposes them to various cyber threats. These threats range from unauthorized access to vehicle systems, manipulation of sensor data, and the injection of false messages into the network (Cao, Li et al., 2023; Kerrache, Calafate, Cano, Lagraa, & Manzoni, 2016). In 2020, a company named Context Information Security discovered that the tire pressure monitoring system in a Ford model could be intercepted, and attackers could send false messages indicating proper inflation despite the tires being deflated. Moreover, if a vehicle is controlled by attackers, it can send malicious messages to conceal traffic events like accidents and traffic jams. These false messages could lead to dire outcomes by misdirecting traffic management systems, precipitating accidents, or even compromising the safety of road users. Therefore, developing robust and efficient techniques to detect and mitigate false messages is vital for ensuring the reliability and integrity of IoV.

False messages refer to any fabricated or malicious information that is injected into the IoV, with the intention to deceive or disrupt its functioning. As shown in Fig. 1, two types of messages are commonly considered in the IoV environments (Chen et al., 2017), i.e., the Basic Safety Message (BSM) and the Event Report Message (ERM). On the one hand, BSMs are beacon messages sent periodically by vehicles, which usually include vehicle coordinates, speed and other information. Malicious vehicles can disrupt traffic monitoring by broadcasting BSMs containing falsified data. Also, as shown in Fig. 1(a), attackers can send lowered speed data and false locations, which may cause consecutive vehicles to falsely slow down and decrease traffic efficiency. On the other hand, ERMs are sent when a vehicle perceives a traffic event to notify other network participants. As shown in Fig. 1(b), malicious

vehicles can cover up existing events through false ERMs reporting that no event has occurred, disrupting applications such as traffic jam avoidance. Additionally, malicious vehicles can send false ERMs reporting non-existing events, causing other vehicles to divert unnecessarily. It is worth noting that the terms ERM and BSM are used for the convenience of representation, since standardization of ITS applications varies by areas in the world. In Europe, the European Telecommunications Standards Institute (ETSI) has specified Cooperative Awareness Message (CAM) in ETSI EN 302 637-2 and Decentralized Environmental Notification Message (DENM) in ETSI EN 302 637-3. According to the specifications, a CAM contains the status information of a vehicle such as position and motion state, while a DENM contains information related to a road hazard or an abnormal traffic conditions. In China, the corresponding standards are developed by the China Society of Automotive Engineers (CSAE). The BSM, which is in line with the CAM in European standards, is defined as one of the Day 1 message frames in T/CSAE 53-2020. In T/CASE 157-2020, the Msg_SSM is defined to share sensor information of obstacles and traffic events, which is similar to the DENM.

The main objective of our work is to accurately identify false messages and malicious vehicles, ensuring the integrity of information disseminated in the IoV. Various research efforts have been made to tackle the issue of false message detection in the IoV. For false BSM detections, existing works generally adopt data mining techniques in both distributed (Zaidi, Milojevic, Rakocevic, Nallanathan, & Rajarajan, 2016) and centralized (Ercan, Ayaida, & Messai, 2021) schemes. In distributed schemes, each vehicle receives and stores BSMs from its neighboring vehicles, and utilizes statistical methods such as hypothesis testing to determine the validity of messages. In centralized schemes, a central server first gathers traffic data from vehicles. Then, machine learning models are trained on these data and used to classify malicious messages. Moreover, multiple BSMs can be reconstructed as trajectory data, on which clustering algorithms are applied to detect anomalous trajectories (Huang, Feng, & Liu, 2021; Wang et al., 2020). For false ERM detections, geospatial information is commonly explored (Zhang, Li, Hou & Shi, 2021). Schemes based on geospatial information assume that reports sent by vehicles closer to the location and time of traffic events are more credible. These reports are assigned higher weights during report aggregation and decision-making. Moreover, blockchain (Khalid et al., 2021) and consensus algorithms (Li, Tseng, Higuchi, Ucar, & Altintas, 2021; Liu, Lin, Kang, Shiraishi, & Blough, 2019) are also employed to validate the status of traffic events. While producing robust results against attackers, consensus algorithms can be heavy on bandwidth consumption and introduce non-trivial consensus delay (Tang, Wang, Jiang, Ge, & Tan, 2022). Additionally, cryptographic techniques provide secure communication channels and authentication mechanisms to prevent message tampering (Lu, Qu, & Liu, 2019).

Despite the progress made, several research challenges remain in the field of false message detection in the IoV. Firstly, the occurrence and development of traffic events are usually relatively rapid. The introduction of security mechanisms must take into account the real-time nature of application scenarios. Validating traffic events through consensus mechanisms may cause an unacceptable delay. Also, the impact of different types of traffic events on traffic flows may be varied. Secondly, the dynamic nature of IoV environments poses a challenge in accurately modeling normal vehicle behaviors, as well as distinguishing them from malicious activities. While data mining techniques prevail in this regard, supervised methods suffer from the lack of training data (especially malicious data), and unsupervised methods may produce inaccurate results.

In light of the above challenges, we propose a detection scheme for false ERMs and BSMs, which leverages the fact that traffic events may influence vehicle movements (e.g. slowing down or crowding around the scene of an event). The main contributions of our work can be summarized as follows:

1. Existing literature seldom adapts to the varied impact of different traffic events on traffic operations. To address this adaptability, a traffic anomaly detection scheme targeting false ERMs is proposed, which quantifies the impact of a traffic event and reduces the potential delay caused by the vehicle consensus process. The vehicle density in an area is treated as time series data, and a time series prediction model is designed to predict densities under normal circumstances. Then, the anomaly index is calculated based on the divergence between predicted and observed data. If the traffic flow is heavily impacted by a traffic event, the event can be predicted with high confidence, and false messages are identified accordingly. In addition, considering the time constraints of traffic event validation, the consensus mechanism is triggered only in situations where the traffic anomaly is insignificant, which avoids the time delay of consensus algorithms.
2. A cluster quality indicator is proposed and integrated with the clustering algorithm for false BSM detections. The indicator is designed concerning two factors: the distance of a clustered data point to its centroid, and the probability distribution of data in two clusters (i.e., malicious and benign). Firstly, the distance of a clustered data point to its centroid is measured with the Cosine Similarity. Secondly, the hypothesis test is employed to determine the probability that BSMs sent by malicious and benign vehicles follow the same distribution. Then, the quality indicator is calculated for the reputation mechanism to update the long-term vehicle evaluations, optimizing the accuracy of unsupervised clustering algorithms.

The remainder of this paper is organized as follows. The related work is presented in Section 2. The system model and preliminaries of our proposed scheme are described in Section 3. The technical details are further introduced in Section 4. Simulation-based evaluations and result analysis are then presented in Section 5. Finally, Section 6 concludes this paper.

2. Related work

2.1. Applications of IoV

By integrating smart technologies into transportation systems, IoV aims to enhance safety, efficiency, and connectivity. The V2V communication enables vehicles to exchange real-time information with nearby vehicles. This facilitates collision avoidance,

traffic flow optimization, and cooperative driving. Moreover, the V2R communication involves interactions between vehicles and road infrastructure (such as traffic lights, signs, and toll booths), which enables dynamic traffic management, adaptive traffic signal control, and efficient toll collection. For example, smart traffic lights can adjust their timing based on real-time traffic conditions, reducing congestion and improving overall traffic flow. Additionally, vehicles can also communicate with cloud network. For instance, when a traffic accident occurs, vehicles can automatically upload critical data (such as accident location and casualty situation) to a central information system via IoV. This allows traffic police and ambulance personnel to provide timely support and rescue services.

To implement these applications, IoV generates massive amounts of data from sensors, cameras, and communication modules, and many applications are heavy on computation power. In this regard, task offloading becomes a crucial technology in IoV (Chen et al., 2023; Cui et al., 2020; Zhang, Dong et al., 2024; Zhang, Piao, Zhang, Zhang & Dong, 2022). By offloading some computation to Mobile Edge Computing (MEC) servers (Zhang, Cao et al., 2022; Zhang, Ni et al., 2022; Zhang, Wang, Zhang, Zhang, Du et al., 2023; Zhang, Zhang et al., 2024), vehicles can reduce their own computational load and improve real-time responsiveness. Moreover, certain IoV tasks, such as accident warnings, demand real-time processing and data integrity, which requires efficient message routing (Liu, Zhang, Liu, Zhang and Wu, 2020; Liu et al., 2021; Zhang, Liu, Liu, Zhang and Cui, 2018; Zhang, Wang, Zhang, Zhang, Yang et al., 2023) and security mechanisms (Su et al., 2021; Zhang, Gao, Liu, Zhang & Zhao, 2019; Zhang, Wang, Song & Zhao, 2014). Additionally, devices in V2X scenarios may not have access to adequate energy and computation resources, which should be taken into consideration when designing networking mechanism for V2X (Zhang, Chen, Cui & Zhang, 2021; Zhang, Li, Zheng, Ming and Pan, 2014; Zhang, Liu, Zhang, & Liang, 2017; Zhang, Ni et al., 2022; Zhang, Zhou & Tang, 2018).

2.2. False ERM detection

The spatio-temporal information can be considered when validating traffic event reports. For instance, Huang, Ruj, Cavenaghi, Stojmenovic, and Nayak (2014) proposed a voting mechanism with variable weights according to the number of vehicles between the event site and message senders. The vehicle that directly observes the event has the highest weight on the messages it sends, while vehicles with longer hops away are assigned smaller weights. The work (Zhang, Li et al., 2021) adopts the Manhattan distance when calculating the distance from the location of a vehicle to an event. This scheme also considers the number of obstacles on the road to the event location in a more realistic scenario. Besides the time and spatial factors, the work (Lv et al., 2022) integrates contact frequency and trajectory similarity to measure the social relations between vehicles, forming a relatively stable metric for malicious vehicle detections.

Leveraging data collected through the IoV, traffic incident prediction schemes can be integrated in false ERM detections. Yu, Zeng, Xue, and Ma (2022) proposed an LSTM-based traffic incident classifier trained on time series of traffic parameters including speed, acceleration, and position. The decision of false messages is then made according to the prediction results. Similarly, Gyawali, Qian, and Hu (2020) utilized the Random Forest model trained on traffic parameters such as average speed to prediction traffic incidents. Moreover, Gokasar, Timurogullari, Özkan, Deveci, and Lv (2023) proposed a modified standard normal deviation based scheme for traffic incident prediction. The Logistic Regression algorithm is utilized to form the variable for deviation calculation, which considers multiple traffic parameters such as speed, density, and flow.

Consensus algorithms are also explored to validate the status of traffic events. Liu et al. (2019) proposed Proof-of-Eligibility, to prevent vehicles outside the event area from participating in the consensus process. Vehicles are pre-installed with a question dataset. When an event report is generated, a series of questions are selected from the question dataset. Questions fall into three types: view, knowledge, and consistency. The main purpose of these questions is to ensure that vehicles have some information about the event, and can perceive the surrounding environment (e.g., the color of nearby buildings). Vehicles may have to rely on on-board sensors to answer these questions. The work (Kumar, Vishwakarma, & Das, 2023) takes advantage of blockchain to construct a decentralized infrastructure. In order to reduce consensus latency and block congestion, the proposed scheme calculates the reputation of vehicles to determine the leader, secondary, and candidate vehicles. The block is only added to the blockchain by the leader, while the secondary vehicles are responsible for validating it. Khalid et al. (2021) presented a traffic event validation system based on blockchain and smart contracts, where an incentive scheme is designed to reward vehicles that perform event validation. Also, a reputation management scheme is proposed to discourage vehicles from sending false messages.

Focusing on autonomous driving applications, Feng, Xu, Zhu, Klaine, and Zhang (2023) proposed a wireless distributed consensus framework, in which the communication links are less reliable. The consensus algorithm is designed based on the Practical Byzantine Fault Tolerance (PBFT) protocol. A veto-collection phase is added prior to the normal consensus process, where the feasibility of consensus proposal is verified. Furthermore, a dynamic grouping protocol is designed to tackle the issue of vehicles leaving and joining the consensus network. Considering a potentially unreliable but synchronous (i.e., if messages are not lost, they are delivered within a limited delay) communication scenario, the work (Li et al., 2021) relies on an approximate consensus to derive the binary consensus result. In approximate consensus, non-faulty vehicles are guaranteed to converge even if a few messages are lost.

2.3. False BSM detection

BSMs with deliberately falsified data can cause severe damage to traffic dynamics. In the work (Wang, Yu, Wu, Wang, & He, 2019), the moving status of vehicles is modeled by the Intelligent Driver Model, while the spreading of malicious messages is modeled using a classical epidemic model. The impact of message spreading is verified by simulations under a single-lane platoon scenario, highlighting the need for the integrity assurance of BSMs. Traffic modeling is also explored in false BSM detections. Liu,

Yang, Zhang and Yang (2020) proposed a false message detection scheme under a traffic accident scenario, which uses the traffic flow theory to analyze vehicular behavior. In their settings, the road capacity is reduced by blocked lanes at an accident site. Thus, two typical traffic patterns are modeled, i.e., lane-blocking and blocking-free. A Bayesian theorem-based method is used to calculate the likelihood for each traffic pattern, and the actual traffic condition is estimated to determine whether the reported accident has actually occurred.

Aiming at V2X applications such as autonomous driving and vehicle platoon, Ju, Zhang, and Tan (2020) adopted a novel simple distributed Kalman filter with a modified generalized likelihood ratio to identify sensor deception attacks in platoon-based connected vehicles. Similarly, Zhao, Gill, Pisu, and Comert (2022) employed the particle filter in the data fusion of cooperative localization to eliminate the impact of false messages. They also leveraged the fault signature table and developed a unique attack detection scheme to isolate false data injection attacks in V2X applications.

Machine learning and statistical schemes are commonly adopted in false BSM detections. Zaidi et al. (2016) proposed an intrusion detection system (IDS) for the IoV based on hypothesis test. The Greenshields model is incorporated to calculate traffic parameters, and the traffic flow is estimated with the Gaussian distribution. Then, the t-test is applied to determine whether vehicles are sending malicious data. In this scheme, each vehicle collects traffic parameters from other vehicles in its vicinity, and monitors them in a distributed manner. Ghaleb et al. (2019) proposed a distributed misbehavior detection scheme based on ensemble learning. The scheme employs the Kalman filter to predict the mobility information of neighboring vehicles. Three types of features are extracted: consistency-based, plausibility-based, and behavioral features. These feature representations are then used as the input to an ensemble learning-based algorithm. Ilango, Ma, and Su (2022) introduced a scheme based on the Autoencoder and Random Forest to detect BSMs with false position data. The scheme also includes a module to identify novel attack signatures and send related samples to the cloud for model retraining. Mongelli (2021) proposed a falsification detection scheme leveraging boolean rules inferred under a data-driven approach, with the aim of designing a human-understandable detection model.

Le and Maple (2019) proposed trajectory-oriented features for false BSM detection. In this scheme, BSMs collected globally are first concatenated into vehicle trajectories. Then, legitimate trajectories are maintained as a dataset, and trajectory distances are designed as features for classification models. Besides trajectory features, information from the physical layer is also utilized. The work (Ercan et al., 2021) utilized the received signal strength and the logarithmic path loss model to estimate the distance between two communicating vehicles. These data complement the commonly used features such as vehicle positions.

2.4. Summary

IoV aims to enhance safety, efficiency, and connectivity in transportation systems, which generates massive amounts of data and requires significant computational power. While task offloading to MEC servers can reduce computational load and improve real-time responsiveness, IoV is also confronted by false messages attacks, especially false ERMs and false BSMs.

False ERM detection involves validating traffic event reports. Spatio-temporal information, such as distance from the event site to message senders, time elapsed since message sending, and obstacles on the road, can be considered for detection. Additionally, social relations between vehicles, contact frequency, and trajectory similarity can aid in distinguishing false messages. These schemes rely heavily on the interaction of communicating vehicles, which may encounter data sparsity issues. Moreover, consensus algorithms and blockchains are explored to validate the status of traffic events and discourage false reporting. However, these schemes may cause a considerable time delay, rendering them unsuitable for time-sensitive traffic events.

Regarding false BSM detection, techniques such as traffic modeling and distributed Kalman filters are utilized, which may be specialized in certain scenarios and have generalization issues. Moreover, machine learning based classifications are also employed to identify malicious messages. These techniques require constantly collecting attack and normal data from individual vehicles, and may need to retrain models for evolving scenarios.

3. System model

This section first describes the network entities and the proposed communication scenario. Then, the threat models considered in our scheme are specified. Additionally, technical preliminaries of our proposed scheme are introduced. The notations used in this paper and their definitions are listed in Table 1.

3.1. Network model

Vehicles, RSUs and the central server are the three main entities in the network. Their functionalities are described as follows:

1. **Vehicle:** Each vehicle is equipped with an On-board Unit (OBU) as the processing unit of wireless communication and event perception. Especially, with OBUs compatible with the Cellular V2X (C-V2X) technology (García-Roger, González, Martín-Sacristán, & Monserrat, 2020), vehicles are capable of establishing communications with other entities through two interfaces, namely PC5 and Uu (García et al., 2021). The PC5 interface is employed in device-to-device communications, which offers low latency and can operate without the coverage of cellular infrastructures. The Uu interface is utilized in network communications (i.e., V2N), providing unicast services over longer distances. Moreover, vehicles can perceive traffic events with on-board sensors, such as cameras and LiDAR systems (Li & Ibanez-Guzman, 2020).

Table 1
Notations & Definitions.

Notation	Definition
vehicle i	The identity of a vehicle
BSM_i^t	The basic safety message sent by vehicle i at time t
ERM_i^t	The event report message sent by vehicle i at time t
d_{rgId}^t	The density value of region $rgId$ at time t
$d_i^t[rgId]$	The perpendicular distance of vehicle i to the border of region $rgId$
$D_{rgId}^t, M_{rgId}^t, MN_{rgId}^t$	The calculated features for region $rgId$
$Pred_{rgId}^t$	The predicted density values
Obs_{rgId}^t	The density values observed by the RSU
$D_{JS}(P \parallel Q)$	The Jensen–Shannon divergence of distributions P and Q
$D_{KL}(P \parallel Q)$	The Kullback–Leibler divergence of distributions P and Q
η	The threshold for anomaly index
EV_i	The reputation evidence stored by the central server for vehicle i
$maliciousSet, benignSet$	Sets of identified malicious and benign vehicles
dis_i	The distance indicator of vehicle i to its cluster centroid
w_i	The weight of each reputation update for vehicle i
rep_i	The reputation of vehicle i
ξ	The reputation threshold

- Road Side Unit (RSU):** RSUs are transportation infrastructures deployed along the road, which take a crucial role in various IoV applications such as event alerts and traffic management. RSUs can also communicate through both the PC5 and Uu interfaces. Since RSUs have relatively higher computation power and storage capacity, sophisticated detection algorithms can be deployed on them.
- The central server:** On the one hand, the central server acts as the certificate authority of IoV, which is responsible for the generation, verification, and revocation of certificates. On the other hand, the central server is the security manager in the network, tracking the long-term behaviors of vehicles with a reputation mechanism. Also, the central server can establish communications within a wide area using cellular technologies.

Fig. 2 shows the architecture of proposed scenario. The map is segmented into multiple rectangular regions according to a pre-defined size, and network participants can calculate the index of region (i.e., $rgId$) they are currently in according to their positions. Given by BSM_i^t , the BSM sent by vehicle i at time t is defined as:

$$BSM_i^t = (vx_i^t, vy_i^t, posx_i^t, posy_i^t), \quad (1)$$

where vx_i^t and vy_i^t refer to the speed of vehicle i in latitude and longitude directions at time t , respectively. $posx_i^t$ and $posy_i^t$ refer to the coordinates of vehicle i at time t , respectively. Moreover, a traffic event occurs at time t in the scenario. Vehicles that perceive this event will broadcast an ERM. Given by ERM_i^t , the ERM sent by vehicle i at time t is defined as:

$$ERM_i^t = (STATUS, t, rgId, vehId_i), \quad (2)$$

where $STATUS \in \{EVENT, CLEAR\}$ indicates that an event occurs or is cleared, and $vehId_i$ is the identity of vehicle i . Upon perceiving the event, vehicles react to the event and reduce their speeds, resulting in an increased vehicle density in the region where the event occurs.

3.2. Threat model

Two threat models are considered in the proposed scenario, namely the false BSM attack and the false ERM attack.

- False BSM attack:** Attackers manipulate the speed and position data sent in BSMs. Given by $FBSM_i^t$, the false BSM sent by vehicle i at time t is specified as:

$$FBSM_i^t = (vx_i^t/2, vy_i^t/2, posx_i^t - a, posy_i^t - a), \quad (3)$$

where a is the offset added to coordinates, and the speeds are halved in the false BSM.

- False ERM attack:** Attackers exhibit two types of malicious behaviors. First, attackers intend to conceal an event that actually occurs. Upon perceiving events, attackers will send ERM with $STATUS$ set to $CLEAR$. Second, attackers can forge a non-existent event by sending ERMs at an arbitrary time and position. The $STATUS$ arguments in these ERMs are set to $EVENT$. However, the $vehId$ cannot be forged, since the Public Key Infrastructure (PKI) is deployed in the network, and the authenticity of message senders can be verified.

3.3. Preliminaries

As shown in Fig. 2, the proposed scheme consists of three functional modules: traffic anomaly detection, consensus mechanism, and BSM clustering. Firstly, as a robust prediction model, the Gradient Boosting Decision Tree (GBDT) is utilized in our scheme

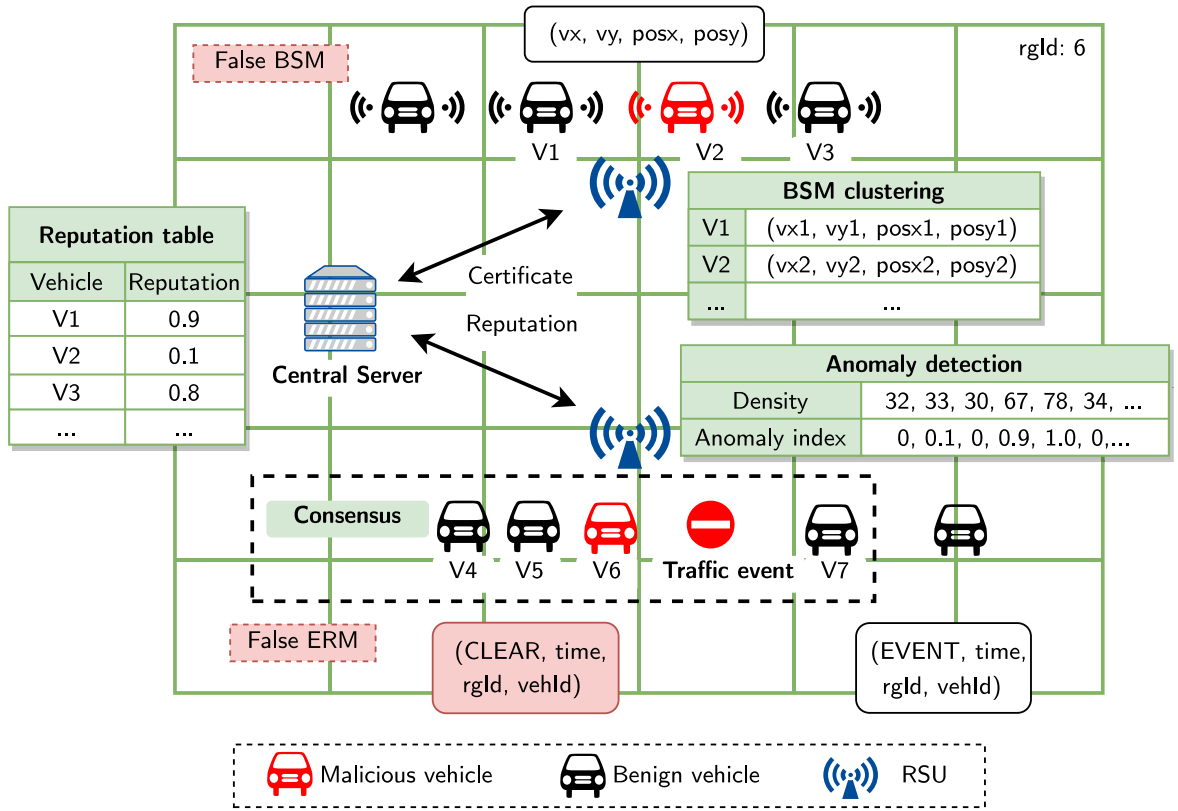


Fig. 2. The architecture of proposed scenario.

for anomaly detection. Secondly, the PBFT is a widely adopted consensus algorithm, which serves as the basis of proposed vehicle consensus mechanism. Thirdly, the BSM clustering employs the K-means++ algorithm, which provides a more reasonable cluster centroid initialization. These three models are detailed in this section below.

3.3.1. Gradient boosting decision tree

The GBDT is a type of ensemble model that combines multiple decision trees to create a strong predictive model. The GBDT model is based on the principle of boosting, which is a technique used to improve the accuracy of weak learners by combining them into a strong learner. The boosting process involves iteratively adding weak learners to the model, with each new learner focusing on the errors made by the previous learners.

The GBDT model employs decision trees as weak learners. Decision trees are a type of supervised learning algorithm applicable to both classification and regression tasks. Such learning algorithms recursively split the data into smaller subsets based on the values of input features, with each split creating a new node in the tree. The final prediction is based on the leaf node that the data point ultimately resides in.

The training of GBDT models starts by creating a single decision tree using the entire data set. Then, the residuals of the first tree are computed, which represent the difference between the actual values and the predicted values of the target variable. After that, the second tree is trained using the residuals as the target variable. This process is repeated for a specified number of trees, with each new tree focusing on the errors made by the trees trained previously. The final prediction is computed by adding up the predictions of all decision trees weighted by their accuracy.

The GBDT model utilizes the gradient descent algorithm to minimize the loss function. The loss function $L(\theta)$ measures the error between the predicted values and the actual values of the target variable, which is calculated as:

$$L(\theta) = \sum_{i=1}^N l(y_i, F_{M-1}(x_i) + \theta h(x_i)), \quad (4)$$

where y_i is the actual value of sample i , N is the number of samples in the training set, M is the number of trees in the ensemble, and θ is the learning rate. $F_{M-1}(x_i)$ is the predicted value of sample i by the first $M-1$ trees in the ensemble, and $h(x_i)$ is the predicted value of sample i by the M th tree. The function $l(\cdot)$ calculates the loss of each tree, which differs based on the task (e.g. Mean Square Error for regression, and Log Likelihood for classification).

In our scheme, the LightGBM (Ke et al., 2017) is chosen as the implementation of GBDT, which offers several advantages over other frameworks, including:

1. **Gradient-based One-Side Sampling:** The LightGBM employs a novel sampling strategy that selects the instances with larger gradients during the training process. This approach focuses on the challenging instances that are more likely to be anomalous, leading to improved model performance.
2. **Leaf-Wise Tree Growth:** Unlike traditional GBDT implementations that grow trees level by level, the LightGBM grows trees in a leaf-wise manner. This strategy reduces the number of tree nodes and improves computational efficiency, making it highly scalable for large datasets.
3. **Regularization Techniques:** The LightGBM incorporates various regularization techniques, such as leaf-wise tree regularization and feature sub-sampling, to prevent overfitting and enhance generalization capabilities.

3.3.2. Practical Byzantine fault tolerance

The PBFT consensus protocol is a fundamental building block in distributed systems. Inspired by the Byzantine Generals' Problem, the PBFT protocol provides a practical solution that enables a group of nodes to reach a consensus despite the presence of Byzantine faults. Byzantine faults refer to arbitrary and malicious behavior (e.g., sending contradictory messages, delaying or dropping messages) exhibited by part of nodes in a distributed system. The PBFT protocol provides a robust and efficient solution to address this challenging problem and has been widely adopted in various scenarios including the IoV.

The PBFT consensus protocol operates in a distributed system comprising a group of nodes, referred to as replicas, that work together to reach an agreement on a sequence of transactions. The protocol can be divided into five distinct phases, each serving a specific purpose in achieving consensus. As shown in Fig. 3, these phases include the request, pre-prepare, prepare, commit, and reply phases. The process in Fig. 3 consists of four replicas, in which Replica 0 is the current primary replica, and Replica 3 is a malicious one. Let f be the number of malicious nodes. The five phases are described as follows:

1. **Request Phase:** The protocol begins when a client sends a request to the primary replica, which acts as the leader. The request contains the desired operation that the client wants to execute.
2. **Pre-prepare Phase:** Upon receiving the client's request, the primary replica assigns a sequence number to the request and broadcasts a pre-prepare message to all other replicas. This message includes the sequence number, the operation, and a hash of the operation.
3. **Prepare Phase:** Upon receiving the pre-prepare message, all replicas verify the message's validity and then broadcast a prepare message to the other replicas. This message includes the sequence number, the hash of the operation, and a signature from the replica. Replicas wait for a threshold number of prepare messages before proceeding to the next phase.
4. **Commit Phase:** Once a replica receives at least $2f + 1$ consistent prepare messages for a particular sequence number, it broadcasts a commit message to all other replicas. This message includes the sequence number, the hash of the operation, and a signature from the replica. Replicas wait for a threshold number of commit messages before proceeding to the next phase.
5. **Reply Phase:** After receiving at least $2f + 1$ consistent commit messages, replicas can execute the requested operation and reply to the client with the result. The client waits for a threshold number of matching replies before considering the operation is completed.

Throughout the protocol, replicas maintain a state machine that tracks the sequence of executed operations. This state machine ensures that all replicas reach the same state, despite the presence of Byzantine faults. By following these phases and exchanging messages, the PBFT protocol ensures that with f malicious nodes and at least $3f + 1$ nodes in total, consensus can be reached.

3.3.3. The K-means++ algorithm

Data clustering is a fundamental task in unsupervised machine learning, aiming to group similar data points together based on their intrinsic properties. Among the various clustering algorithms, the K-means algorithm has been widely adopted due to its simplicity and efficiency. However, the traditional K-means algorithm is highly sensitive to the initial selection of cluster centroids, often leading to suboptimal solutions. To address this limitation, the K-means++ algorithm was proposed as an enhancement, providing a more robust and effective initialization scheme. The principle behind K-means++ is to choose initial centroids that are far apart from each other, increasing the likelihood of finding better cluster assignments. The algorithm assigns a higher probability to data points that are farther away from the existing centroids, ensuring a more even distribution of initial centroids across the dataset.

The K-means++ algorithm consists of the following steps:

- Step 1. **Initialization.** The algorithm randomly selects the first centroid from the dataset. This initial centroid can be chosen uniformly at random or based on a specific criterion.
- Step 2. **Distance calculation.** For each data point, the algorithm calculates the minimum squared Euclidean distance to the nearest centroid. This distance serves as a measure of how far a data point is from the existing centroids, which can be customized according to different applications.
- Step 3. **Centroid selection.** The next centroid is chosen based on a weighted probability distribution, where data points with larger distances have higher probabilities of being selected as centroids. The probability of selecting a data point as the next centroid is proportional to its squared distance from the nearest centroid.

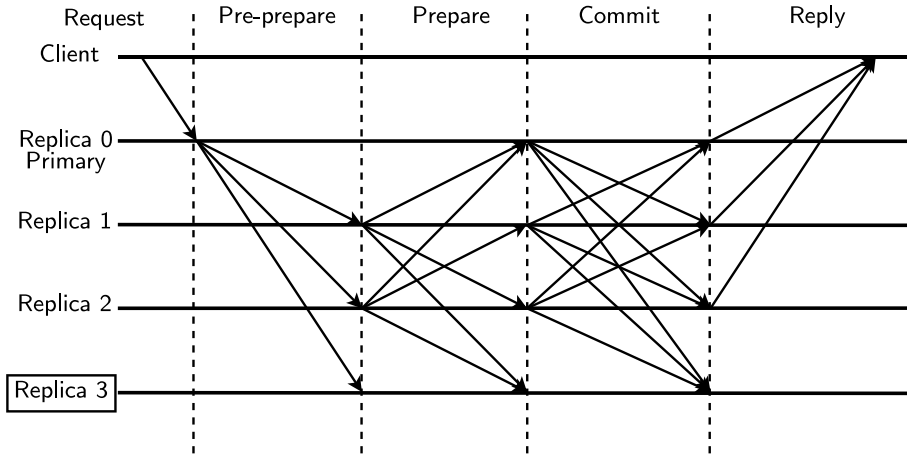


Fig. 3. The process of PBFT consensus.

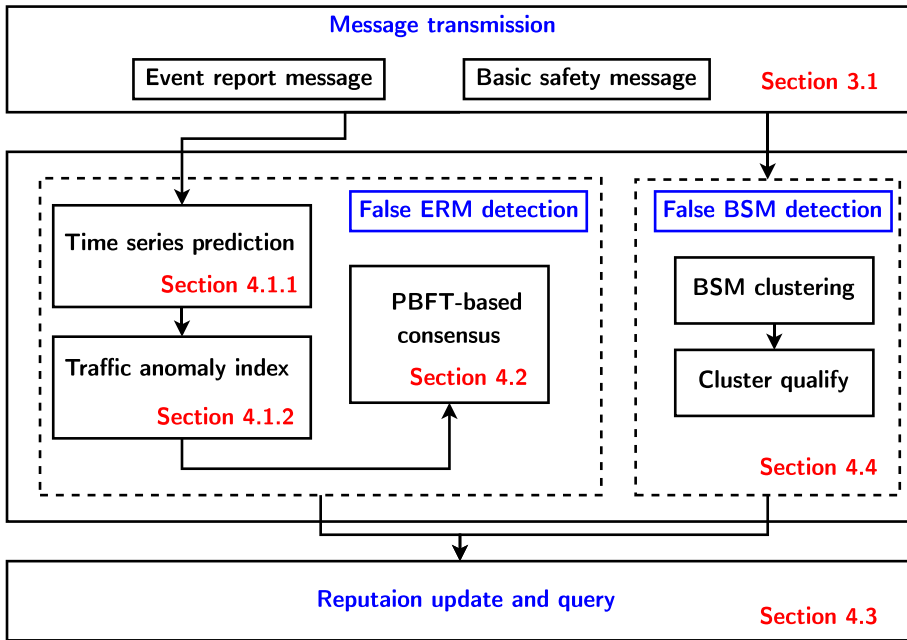


Fig. 4. Overview of the proposed scheme.

Step 4. **Repeat Steps 2 and 3.** Steps 2 and 3 are repeated until the desired number of centroids is obtained.

Step 5. **K-means clustering.** Once the initial centroids are determined, the K-means algorithm proceeds with the standard clustering process, iteratively updating the cluster assignments and centroid positions until convergence.

The K-means++ algorithm differs from the traditional K-means algorithm primarily in the initialization step. While K-means randomly selects initial centroids, K-means++ employs a more intelligent and systematic approach, selecting initial centroids that are well-distributed across the dataset. This renders K-means++ better robustness and convergence speed compared to K-means.

4. Proposed scheme

As illustrated in Fig. 4, the proposed scheme consists of two mechanisms: traffic event validation and BSM filtering, aiming at false ERM attacks and false BSM attacks, respectively. The traffic event validation mechanism includes two stages, i.e., traffic anomaly detection and vehicle consensus formation. The BSM filtering mechanism mainly adopts a data clustering approach, with the assistance of a cluster qualify indicator. The technical details of these two mechanisms are introduced in this section.

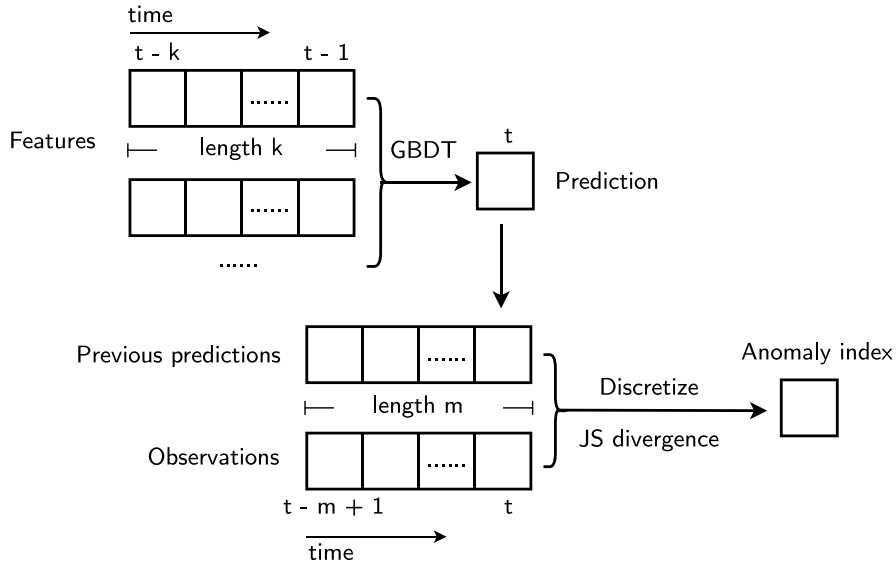


Fig. 5. The process of traffic anomaly detection.

4.1. Traffic anomaly detection

The traffic anomaly detection mechanism aims to capture the unusual pattern of traffic data when a traffic event occurs. Through BSMs periodically sent by vehicles, RSU can derive the vehicle density of regions it covers. In this mechanism, the vehicle density is represented by the number of vehicles in a region, and is treated as time series data. Thus, a time series prediction model is designed to predict the density values under normal traffic conditions in the next time intervals. Then, the divergence between predicted values and real values is calculated. If the divergence is significant enough, it is convincing that an event occurs. The above process is illustrated in Fig. 5 and detailed below.

4.1.1. Feature selection

Three factors are considered when composing features for the prediction model:

1. **Previous density values:** The previous density values reflect the changing pattern over time, and are critical to predicting future values. Therefore, a sliding window is applied to the time series data of density values. To predict the density value at time t , the data points within the window are utilized as the predicting feature, which is denoted by D_{rgId}^t and represented as:

$$D_{rgId}^t = \{d_{rgId}^{t-k}, d_{rgId}^{t-k+1}, \dots, d_{rgId}^{t-2}, d_{rgId}^{t-1}\}, \quad (5)$$

where k is the length of sliding window, and d_{rgId}^{t-k} refers to the density value of region $rgId$ at time $t-k$.

2. **Movement:** The movement of vehicles in a region has a significant influence on vehicle density. Denoted by M_{rgId}^t , this feature is represented as:

$$M_{rgId}^t = \left\{ \frac{1}{|V_{rgId}^{t-1}|} \sum_{i \in V_{rgId}^{t-1}} vx_i^{t-1}, \frac{1}{|V_{rgId}^{t-1}|} \sum_{i \in V_{rgId}^{t-1}} vy_i^{t-1}, \frac{1}{|V_{rgId}^{t-1}|} \sum_{i \in V_{rgId}^{t-1}} posx_i^{t-1}, \frac{1}{|V_{rgId}^{t-1}|} \sum_{i \in V_{rgId}^{t-1}} posy_i^{t-1} \right\}, \quad (6)$$

where V_{rgId}^{t-1} is the set of vehicles in region $rgId$ at time $t-1$, and $|V_{rgId}^{t-1}|$ is the size of V_{rgId}^{t-1} .

3. **Neighboring movement:** The vehicle density of a region is also affected by the movement of vehicles in its neighboring regions. Here, the speed and distance perpendicular to the border of two regions are considered. Vehicles with higher speeds and shorter distances are more likely to affect the density of a neighboring region. For the convenience of expression, the region (denoted by $rgId'$) to the west of region $rgId$ is taken as an example. Let MW_{rgId}^t be the influence factor of region $rgId'$ on region $rgId$ at time t , which is calculated as:

$$MW_{rgId}^t = \frac{1}{|V_{rgId'}^{t-1}|} \sum_{i \in V_{rgId'}^{t-1}} vx_i^{t-1} \cdot \frac{1}{d_i^{t-1}[rgId]}, \quad (7)$$

where $d_i^{t-1}[rgId]$ refers to the perpendicular distance of vehicle i to the border of region $rgId$. Here, since the factor is being calculated for the region on the west side, the speed in the longitude direction (i.e., vx_i^{t-1}) is utilized. Let ME_{rgId}^t , MN_{rgId}^t ,

MS_{rgId}^t be the factors for regions on the east, north, south side of the region $rgId$, respectively. Given by NM_{rgId}^t , this feature is represented as:

$$NM_{rgId}^t = \{MW_{rgId}^t, ME_{rgId}^t, MN_{rgId}^t, MS_{rgId}^t\}. \quad (8)$$

The above features are calculated based on historic traffic data gathered through simulations in circumstances where no attackers and events are present. Then, the GBDT regression model (as specified in Section 3.3.1) is trained to predict the density value at time t . In the context of anomaly detection, GBDT models can effectively capture complex patterns and non-linear relationships present in time series data. However, since the model is trained on the data under normal traffic conditions, it cannot accurately predict the density values of regions affected by traffic events. Therefore, this discrepancy can be calculated to determine whether a traffic event is present in a region.

4.1.2. Anomaly index calculation

At each time step, the RSU that covers region $rgId$ calculates the proposed features, and persists the predicted density value into its local storage. At time t , the RSU extracts the stored predictions for the previous m time intervals, and organizes them into a sequence $Pred_{rgId}^t$. The sequence is represented as:

$$Pred_{rgId}^t = \{\overline{ds}_{rgId}^{t-m+1}, \overline{ds}_{rgId}^{t-m+2}, \dots, \overline{ds}_{rgId}^t\}, \quad (9)$$

where $\overline{ds}_{rgId}^t$ is the predicted density value at time t . Additionally, the RSU also maintains a sequence of the observed density values for the previous m time intervals, which is denoted by Obs_{rgId}^t and represented as:

$$Obs_{rgId}^t = \{ds_{rgId}^{t-m+1}, ds_{rgId}^{t-m+2}, \dots, ds_{rgId}^t\}. \quad (10)$$

Algorithm 1: The anomaly index calculation

Input: $Pred_{rgId}^t$, Obs_{rgId}^t , the number of bins n
Result: The anomaly index

- 1 $S_1 \leftarrow$ concatenate $Pred_{rgId}^t$ and Obs_{rgId}^t ;
- 2 $interval \leftarrow \frac{\max(S_1) - \min(S_1)}{n}$;
- 3 Divide S_1 into n bins with width $interval$;
- 4 $S_2 \leftarrow$ transform $Pred_{rgId}^t$ based on the binning;
- 5 $S_3 \leftarrow$ transform Obs_{rgId}^t based on the binning;
- 6 $P_{rgId}^t \leftarrow$ zero array of size n ;
- 7 $Q_{rgId}^t \leftarrow$ zero array of size n ;
- 8 **for each data point** v **in** S_2 **do**
- 9 $P_{rgId}^t(v) \leftarrow P_{rgId}^t(v) + 1$;
- 10 **end**
- 11 **for each data point** v **in** S_3 **do**
- 12 $Q_{rgId}^t(v) \leftarrow Q_{rgId}^t(v) + 1$;
- 13 **end**
- 14 Normalize P_{rgId}^t and Q_{rgId}^t ;
- 15 $index \leftarrow$ Jensen-Shannon divergence of P_{rgId}^t and Q_{rgId}^t ;
- 16 **return** $index$;

Algorithm 1 details the calculation of anomaly index with the above two sequences as the input. To measure the discrepancy between $Pred_{rgId}^t$ and Obs_{rgId}^t , the two sequences are preprocessed into two discrete probability distributions P_{rgId}^t and Q_{rgId}^t , respectively. Firstly, the equal width binning discretization method is applied to divide a continuous sequence into n equally-sized bins (Line 1-3). Each data point in the sequence is then replaced by the index of bin it falls into (Line 4-5). Secondly, P_{rgId}^t and Q_{rgId}^t are formed according to the discretization result (Line 6-14). Thirdly, the Jensen-Shannon divergence (JS divergence) between P_{rgId}^t and Q_{rgId}^t is denoted by D_{JS} and calculated as:

$$D_{JS}(P_{rgId}^t \parallel Q_{rgId}^t) = \frac{1}{2} \left(D_{KL}(P_{rgId}^t \parallel M_{rgId}^t) + D_{KL}(Q_{rgId}^t \parallel M_{rgId}^t) \right), \quad (11)$$

where D_{KL} is the Kullback-Leibler divergence (KL divergence), and $M_{rgId}^t = \frac{1}{2}(P_{rgId}^t + Q_{rgId}^t)$ is the average of the two distributions. The KL divergence is further calculated as:

$$D_{KL}(P_{rgId}^t \parallel Q_{rgId}^t) = \sum_i P_{rgId}^t(i) \log \left(\frac{P_{rgId}^t(i)}{Q_{rgId}^t(i)} \right), \quad (12)$$

where i is the index of the elements in the distributions. The JS divergence is utilized as the anomaly index, which measures the degree of unusual changes in traffic (Line 15-16). If the anomaly index is higher than a preset threshold η , a traffic event is believed to have occurred.

4.2. Vehicle consensus formation

The vehicle consensus formation process consists of two phases: (i) consensus vehicle selection, and (ii) the consensus algorithm execution among selected vehicles. Two types of network participants are involved in the process, i.e., vehicles and RSUs. Four roles are defined for the participants: {WAITING, CLIENT, REPLICAS, FINISHED}. The consensus formation process is detailed in Algorithm 2, and is described below.

Algorithm 2: Consensus formation

Input: ERM from reporting vehicles
Result: The consensus result

```

1 RSU loop:
2    $Role \leftarrow WAITING$ ;
3   if  $Role$  is WAITING then
4      $vehList \leftarrow$  all reporting vehicles;
5     Query vehicle reputations from the central server;
6     Sort  $vehList$  by reputations;
7      $vehGroup \leftarrow$  selected vehicles with highest reputations;
8     Send SETUP messages to  $vehGroup$ ;
9      $Role \leftarrow CLIENT$ ;
10  else if  $Role$  is CLIENT then
11    Wait for CONFIRM from  $vehGroup$ ;
12    Send REQUEST message to replicas;
13     $Role \leftarrow FINISHED$ ;
14  else if  $Role$  is FINISHED then
15    Wait for REPLY message;
16    Identify reports that are inconsistent with the consensus result;
17     $Role \leftarrow WAITING$ ;
18  end
19 Vehicle loop:
20    $Role \leftarrow WAITING$ ;
21   if  $Role$  is WAITING then
22     Wait for SETUP message from RSU;
23     Send CONFIRM message to RSU;
24     Setup  $peerList$  based on  $vehGroup$ ;
25      $Role \leftarrow REPLICAS$ ;
26   else if  $Role$  is REPLICAS then
27     Wait for REQUEST from client;
28     Run consensus algorithm within  $peerList$ ;
29      $Role \leftarrow FINISHED$ ;
30   else if  $Role$  is FINISHED then
31     Send REPLY to client;
32      $Role \leftarrow WAITING$ ;
33  end

```

- Step 1. Both vehicles and the RSU are in the role of WAITING.
- Step 2. The RSU receives ERM from vehicles in its vicinity, and queries the central server for the reputations of the reporting vehicles. Then, the RSU selects c vehicles with the highest reputations (introduced in Section 4.3), forming the consensus group. The number of selected vehicles c is within $[3, C]$, where C is a preset constant. If the RSU has less than three vehicles in its vicinity, the consensus process is aborted. After that, the RSU sends SETUP messages to vehicles in the consensus group, and transforms its role to CLIENT (Line 3–9). The identities of vehicles in the consensus group are included in the SETUP message.
- Step 3. Upon receiving the SETUP message, vehicles send CONFIRM messages back to the RSU. Then, vehicles set themselves as REPLICAS (Line 21–25).
- Step 4. After receiving all CONFIRM messages, the RSU sends REQUEST messages to the consensus group (i.e. replicas). The REQUEST message includes an operation, which directs vehicles to reach a consensus on the $rgId$ of region where the event occurs. Then, the role of RSU becomes FINISHED (Line 10–13).
- Step 5. Vehicles run the requested operation through the PBFT consensus algorithm, as is specified in Section 3.3.2. Then, the role of vehicles becomes FINISHED (Line 26–29).

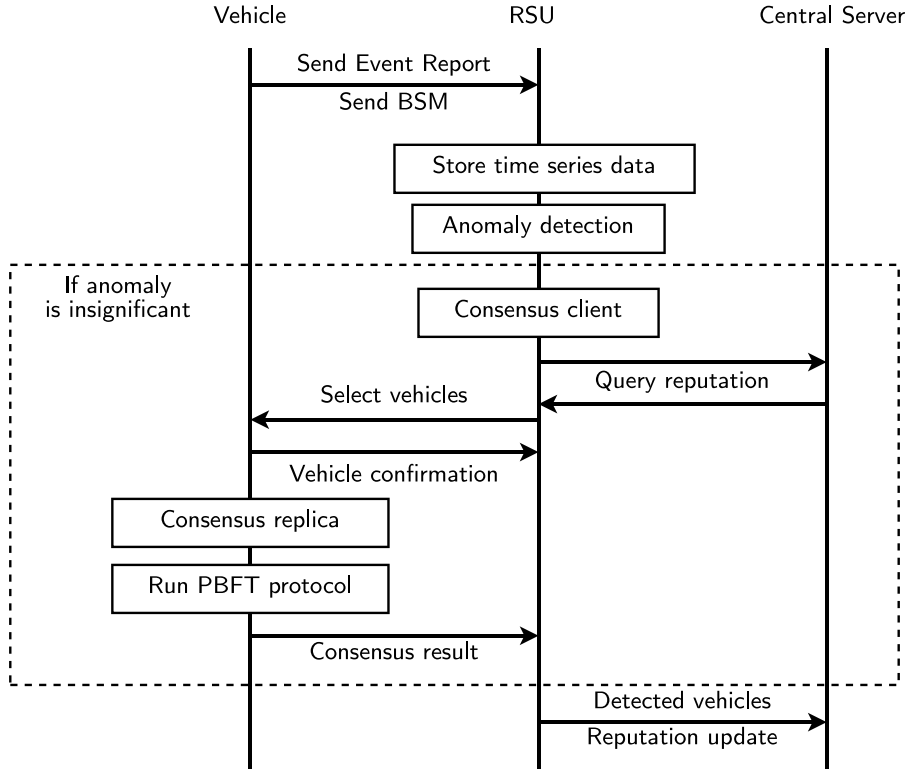


Fig. 6. The process of false event detection.

Step 6. Vehicles send REPLY messages containing the consensus result to the RSU. The RSU then identifies vehicles that send reports inconsistent with the consensus result (Line 14–17, 30–32).

The above algorithm inherits the property about the amount of tolerated faulty nodes from PBFT, which is shown in [Theorem 1](#).

Theorem 1. Let c be the number of selected vehicles running the consensus, and f be the number of vehicles with byzantine failure in the selected vehicles. Then, $3f + 1 \leq c$ has to be satisfied for the correct consensus to be reached.

Proof. Firstly, the f vehicles with byzantine failure may not send any consensus messages. To ensure that the consensus process can proceed in this condition, the consensus has to be reached with $c - f$ messages. Secondly, among the $c - f$ messages, f messages may be sent by vehicles with byzantine failure and contain malicious event status. To reach the correct consensus, the number of messages from benign vehicles has to be greater than the number of messages from vehicles with byzantine failure, i.e., $(c - f) - f > f$. Therefore, $3f + 1 \leq c$ has to be satisfied for the consensus process.

4.3. Traffic event validation

Fig. 6 illustrates the traffic event validation mechanism consisting of the above two mechanisms. The mechanism is triggered when an RSU receives ERM from more than k vehicles. First, the anomaly detection process is performed to calculate the anomaly index of vehicle density. If the anomaly index is higher than the threshold η , the reported event is considered a real event, and detection decisions can be made. In this case, the consensus formation process will not be executed, to minimize the time delay of this mechanism. Second, if the anomaly index is lower than η , the consensus formation process is utilized to further determine the validity of reported event. The detection result is uploaded to the central server by the RSU. The central server maintains a reputation score for each vehicle for long-term security management.

A reputation model based on that proposed in [Xiao and Liu \(2019\)](#) is deployed on the central server. This model is based on the Bayesian Inference theory, and assumes that the prior distribution of reputation values is $\text{beta}(a, b)$. Therefore, the model maintains

an evidence record EV_i for each vehicle i , which is represented as:

$$EV_i = (unsat_i, sat_i, a_i, b_i), \quad (13)$$

where $unsat_i$ is the numbers of times vehicle i is detected as attackers, and sat_i is the numbers of times vehicle i behaves normally. a_i and b_i are two parameters of the *beta* distribution, with the initial value being 1. Then, the reputation of vehicle i is updated according to Algorithm 3. The update weight can be set based on the application scenario (e.g. the emergency of an event). Ultimately, vehicles with a reputation below a preset threshold ξ are identified as malicious.

Algorithm 3: Reputation update

Input: EV_i , detection result of vehicle i , update weight w_i for vehicle i
Result: Updated reputation for vehicle i (i.e., rep_i)

```

1   $unsat_i, sat_i, a_i, b_i \leftarrow EV_i$ ;
2  if vehicle  $i$  is malicious then
3     $unsat_i \leftarrow unsat_i + w_i$ ;
4  else if vehicle  $i$  is normal then
5     $sat_i \leftarrow sat_i + w_i$ ;
6  end
7  if  $a_i = 1$  or  $b_i = 1$  then
8    if  $sat_i + unsat_i < a_i + b_i + 1$  then
9      return;
10   end
11 end
12  $a_i \leftarrow sat_i + a_i$ ;
13  $b_i \leftarrow unsat_i + b_i$ ;
14  $EV_i \leftarrow (unsat_i, sat_i, a_i, b_i)$ ;
15  $rep_i \leftarrow \frac{sat_i + a_i}{sat_i + unsat_i + a_i + b_i}$ ;
16 return  $rep_i$ ;

```

4.4. Complexity of false ERM detection

Let N be the number of data samples of GBDT, and M be the number of trees in the ensemble. Let V be the number of ERM reporting vehicles, and c be the number of selected vehicles running the consensus. Then, the time complexity of false ERM detection is $O(M \log(N)) + O(V \log(V)) + O(c^2)$.

Proof. The false ERM detection mainly consists of the GBDT model and the consensus mechanism. Since the number of features is fixed, the time complexity of querying a GBDT is $O(M \log(N))$. Sorting the V vehicles by their reputation scores requires $O(V \log(V))$, and the PBFT algorithm takes $O(c^2)$. In summary, the time complexity of false ERM detection is $O(M \log(N)) + O(V \log(V)) + O(c^2)$.

4.5. BSM filtering

The BSM filtering mechanism combines the data clustering technique and hypothesis test to identify false BSMs. The main goal of this mechanism is to eliminate the negative effect false BSMs have on traffic data collection applications. As shown in Fig. 7, RSUs periodically execute the filtering mechanism, with BSMs received from vehicles as the input. Considering the numerical discrepancy between false and normal BSMs, clustering algorithms can be employed to separate these two types of messages. The following steps are taken for data clustering:

- Step 1. **Data preparation.** Each BSM from vehicle i is extracted as a data point for clustering, in the format of $(vx_i^t, vy_i^t, posx_i^t, posy_i^t)$.
- Step 2. **Message clustering.** The K-means++ algorithm (specified in Section 3.3.3) is utilized to cluster all the extracted data points into two clusters, referred to as CL_1 and CL_2 . The cost function of K-means++ is represented as:

$$cost(CL_{1:k}, z_{1:k}) = \sum_{j=1}^k \sum_{x \in CL_j} \|x - z_j\|^2 \quad (14)$$

where $CL_{1:k}$ and $z_{1:k}$ denote the k clusters and the corresponding cluster centroids. Here, k is equal to 2.

- Step 3. **Vehicle extraction.** For BSMs in each cluster, the message-sending vehicles are extracted as a vehicle set. Let VS_1 and VS_2 denote the two vehicle sets extracted from CL_1 and CL_2 , where the size of VS_1 is greater than the size of VS_2 . Since it is assumed that the majority of vehicles are benign vehicles, it is highly likely that malicious vehicles are contained in VS_2 . Then, malicious vehicles and benign vehicles are identified as follows:

$$maliciousSet = VS_2 \setminus VS_1, \quad (15)$$

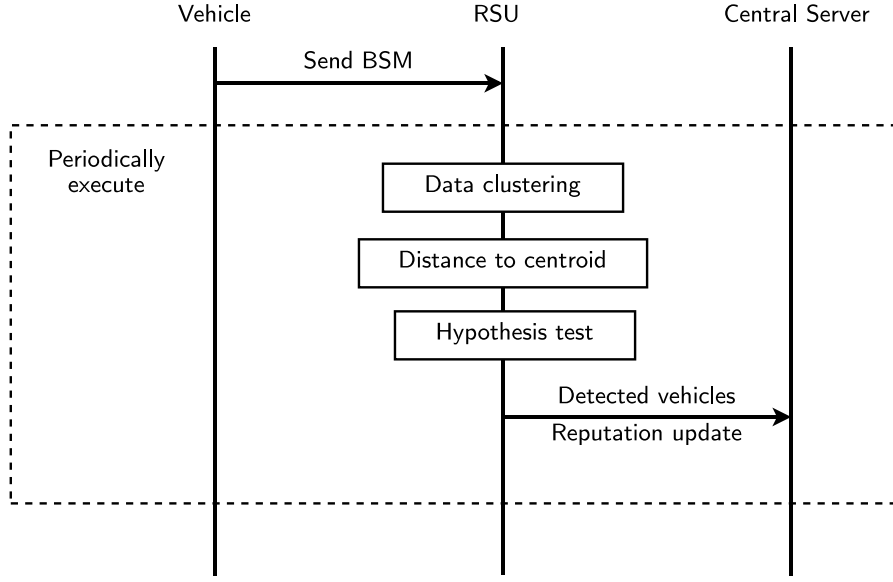


Fig. 7. The process of false BSM detection.

$$\text{benignSet} = VS_1, \quad (16)$$

where $VS_2 \setminus VS_1$ denotes the difference of two sets.

The convergence of above clustering based algorithm is shown in [Theorem 2](#).

Theorem 2. *The cost function decreases during iterations of the clustering algorithm.*

Proof. Let $CL_1^t, \dots, CL_k^t$ denote clusters at iteration t , and $z_1^t, \dots, z_k^t$ denote the centroids of clusters at iteration t . During each iteration, each data point is assigned to its closest centroid, which can be represented as:

$$\text{cost}(CL_{1:k}^{t+1}, z_{1:k}^t) \leq \text{cost}(CL_{1:k}^t, z_{1:k}^t). \quad (17)$$

Then, each centroid is moved to the mean position of data points in the cluster. Since for any cluster CL_i and centroid z_i , the following equation is satisfied:

$$\text{cost}(CL_i, z_i) = \text{cost}(CL_i, \text{mean}(CL_i)) + |CL_i| \cdot \|z_i - \text{mean}(CL_i)\|^2 \leq \text{cost}(CL_i, \text{mean}(CL_i)), \quad (18)$$

the cost function decreases with the repositioned centroids, which can be represented as:

$$\text{cost}(CL_{1:k}^{t+1}, z_{1:k}^{t+1}) \leq \text{cost}(CL_{1:k}^t, z_{1:k}^t). \quad (19)$$

Here, the K-means++ algorithm is chosen to initialize the cluster centroids, which yields superior results compared to the K-means algorithm.

Theorem 3. *Let T be the centroids returned by the K-means++ algorithm, and T^* be the optimal centroids. Then, $E[\text{cost}(T)] \leq O(\log k) \cdot \text{cost}(T^*)$.*

Proof. Let H_t denote optimal clusters at iteration t , and U_t denote the remaining clusters at iteration t . The cost function when k centroids are chosen can be represented as:

$$\text{cost}(T) = \text{cost}(H_k, T_k) + \text{cost}(U_k, T_k) = \text{cost}(H_k, T_k) + \frac{(k - |H_k|) \cdot \text{cost}(U_k, T_k)}{|U_k|}. \quad (20)$$

Then, the mathematical expectation of $cost(T)$ is:

$$\begin{aligned}
E[cost(T)] &= E[cost(H_k, T_k)] + \sum_{t=0}^{k-1} E\left[\frac{(k - |H_{t+1}|) \cdot cost(U_{t+1}, T_{t+1})}{|U_{t+1}|} - \frac{(k - |H_t|) \cdot cost(U_t, T_t)}{|U_t|}\right] \\
&\leq E[cost(H_k, T_k)] + \sum_{t=0}^{k-1} E\left[\frac{(k - |H_t| + 1) \cdot cost(U_t, T_t)}{|U_t|} - \frac{(k - |H_t|) \cdot cost(U_t, T_t)}{|U_t|}\right] \\
&= E[cost(H_k, T_k)] + \sum_{t=0}^{k-1} E\left[\frac{cost(U_t, T_t)}{|U_t|}\right] \\
&= E[cost(H_k, T_k)] + \sum_{t=0}^{k-1} \frac{cost(U_t, T_t)}{|U_t|} \cdot \frac{cost(H_t, T_t)}{cost(T_t)} \\
&\leq E[cost(H_k, T_k)] + \sum_{t=0}^{k-1} \frac{cost(H_t, T_t)}{k - t} \\
&\leq 8cost(T^*)(1 + 1 + \frac{1}{2} + \dots + \frac{1}{k}) \\
&\leq O(logk) \cdot cost(T^*).
\end{aligned} \tag{21}$$

Given the clustering result, the same reputation model as introduced in Section 4.3 is also applied here. The update weight in Algorithm 3 is calculated in relation to the quality of clustering. Specifically, the quality is measured by a combination of two indicators: distance to the cluster centroid, and hypothesis test.

First, two new clusters are built based on the separation of *maliciousSet* and *benignSet*. BSMs from vehicles in each set form a new cluster, referred to as CL_m (for malicious vehicles) and CL_b (for benign vehicles). The centroid of a cluster is calculated by averaging the data points in it. Here, the two components of speed (i.e., vx_i^t and vy_i^t) is combined into $v_i^t = \sqrt{(vx_i^t)^2 + (vy_i^t)^2}$. The centroids for CL_m (given by $\overline{CT_m}$) and CL_b (given by $\overline{CT_b}$) are represented as:

$$\overline{CT_m} = (\frac{1}{|CL_m|} \sum_{i \in \text{maliciousSet}} \sum_t v_i^t, \frac{1}{|CL_m|} \sum_{i \in \text{maliciousSet}} \sum_t posx_i^t, \frac{1}{|CL_m|} \sum_{i \in \text{maliciousSet}} \sum_t posy_i^t), \tag{22}$$

$$\overline{CT_b} = (\frac{1}{|CL_b|} \sum_{i \in \text{benignSet}} \sum_t v_i^t, \frac{1}{|CL_b|} \sum_{i \in \text{benignSet}} \sum_t posx_i^t, \frac{1}{|CL_b|} \sum_{i \in \text{benignSet}} \sum_t posy_i^t). \tag{23}$$

Then, given by dis_i^t , the distance of each data point to its cluster centroid is measured by the Cosine Similarity, calculated as:

$$dis_i^t = \frac{\overline{CT_m} \cdot \overline{dp_i^t}}{\|\overline{CT_m}\| \cdot \|\overline{dp_i^t}\|}, \tag{24}$$

where $\overline{dp_i^t} = (v_i^t, posx_i^t, posy_i^t)$, and the notation $\|\overline{CT_m}\|$ means the modulus of vector $\overline{CT_m}$. Finally, the distance indicator for vehicle i (referred to as dis_i) is calculated by averaging all dis_i^t in terms of time t .

Furthermore, the Kolmogorov–Smirnov (KS) hypothesis test is utilized to quantify the dissimilarity between malicious and benign vehicles. Specifically, the two-sample KS test is applied. The null hypothesis is that the two samples are drawn from the same continuous distribution, while the alternative hypothesis suggests that the two samples are drawn from different distributions. The test statistic measures the maximum vertical difference between the empirical Cumulative Distribution Functions (CDF) of two samples. If the test statistic is large and the p -value associated with the test is below a chosen significance level, it indicates that there is evidence to reject the null hypothesis and conclude that the two samples come from different distributions.

Here, the speed v_i^t is taken as the tested dimension. The two samples (given by S_m for *maliciousSet*, and S_b for *benignSet*) are formed as:

$$S_m = \{v_i^t \mid i \in \text{maliciousSet}\}, S_b = \{v_i^t \mid i \in \text{benignSet}\}. \tag{25}$$

Let F_m and F_b denote the CDFs of S_m and S_b . The test statistic D and p -value are calculated as:

$$D = \max_x |F_m(x) - F_b(x)|, \tag{26}$$

$$p = 2e^{-2D^2 \frac{|S_m| \cdot |S_b|}{|S_m| + |S_b|}}. \tag{27}$$

Then, given by w_i , the update weight for vehicle i is calculated as:

$$w_i = 10^b \cdot (1 - p) \cdot dis_i, \tag{28}$$

where b is the smallest integer that ensures w_i is greater than 1.

4.6. Complexity of false BSM detection

Let K be the number of data points, and I be the maximum number of iterations in the clustering process. The time complexity of false BSM detection is $O(K + 2KI)$.

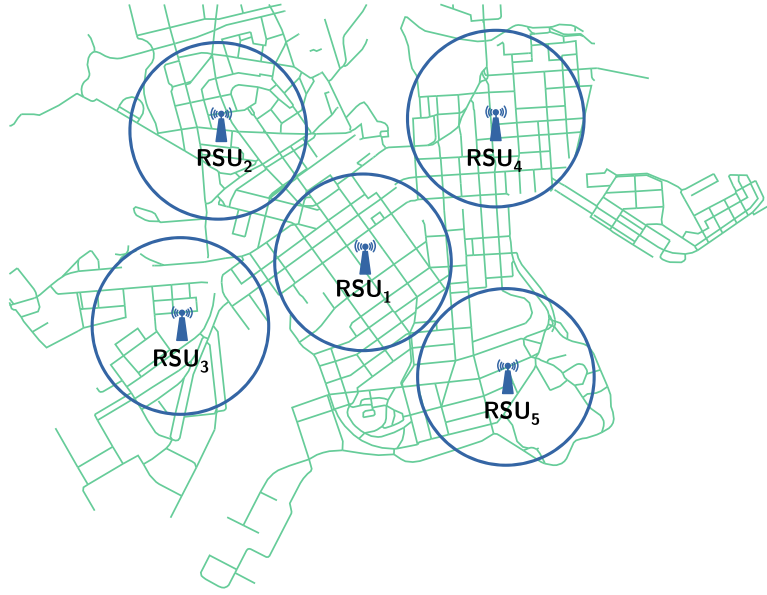


Fig. 8. Deployment of RSUs.

Proof. The time consumption of false BSM detection is dominated by the K-means++ algorithm. The initialization process of K-means++ takes $O(KI)$ time, and the time complexity to form two clusters is $O(KI)$. Then, computing cluster quality takes $O(K)$. Therefore, the time complexity is represented as $O(K + 2KI)$.

5. Performance evaluation

5.1. Simulation setup

The performance of our proposed scheme is evaluated by simulations conducted with the Opportunistic Network Environment simulator (Keränen, Ott, & Kärkkäinen, 2009). The ONE simulator is written in Java based on a discrete event engine and consists of comprehensive functional packages, providing simulations of core network entities, routing algorithms, the application layer, node movement models, etc. Most importantly, for mobility simulations, apart from importing maps and trajectories from external sources, the ONE simulator can also utilize programmed movement models. The commonly used movement models such as random movement, map-based movement, and shortest path movement model have been implemented as utilities. The ONE simulator is commonly used for networking (Song et al., 2023), vehicle scheduling (Hu et al., 2023), and security research (Wang et al., 2023) in IoV. Regarding the anomaly detection model, the Predictive Model Markup Language (PMML) is utilized to simplify the training of GBDT and integrate GBDT with the ONE simulator. The PMML is a standard describing different data mining algorithms, which is implemented in many programming languages such as Java and Python. Therefore, the PMML can be utilized to exchange models between different programming languages. In our experiments, the vehicle movements data generated by the mobility simulation module are exported. Then, vehicle density values are extracted and serve as the training data for GBDT. The model is trained using the LightGBM Python package. After training, the model is exported and converted to PMML using the JPMML-LightGBM package. Lastly, the converted model is loaded in the Java-based simulator with JPMML-Evaluator.

The simulation scenario is taken from the road network of Helsinki city, covering an area of $4500 \times 3400 \text{ m}^2$. This scenario has been used in multiple published papers regarding the security of IoV (Cheong et al., 2024; Wang et al., 2023), and is capable of evaluating the proposed false message detection scheme. Additionally, five RSUs are deployed along the road as shown in Fig. 8. The movements of 150 vehicles are generated based on the shortest path toward predefined destinations. Detailed simulation parameters are listed in Table 2.

To evaluate the performance of detection schemes under different attack intensities, the proportion of malicious vehicles ranges from 10% to 45% in our simulations, with the assumption that the majority of vehicles are benign vehicles. Four metrics are utilized for performance evaluations:

1. **Precision:** the ratio between the number of vehicles detected correctly as malicious and the number of detected vehicles.
2. **Recall:** the ratio between the number of vehicles detected correctly as malicious and the number of malicious vehicles.
3. **F1 score:** the harmonic mean of precision and recall, reflecting the overall performance of a detection scheme. The F1 score is calculated as:

$$F1 = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}.$$

Table 2
Simulation parameters.

Parameter	Value
Simulation time	5400 s
Map size	4500 × 3400 m ²
Total number of vehicles	150
Total number of RSUs	5
Vehicle speed	10–50 km/h
V2V communication range	100 m
Covering radius of RSU	500 m
Bandwidth	500 kbps
Coordinate of the event	(2000, 1500)
Duration of the event	500 s
Covering radius of the event	400 m
η	0.7
k	30
m	20
C	10
ξ	0.5

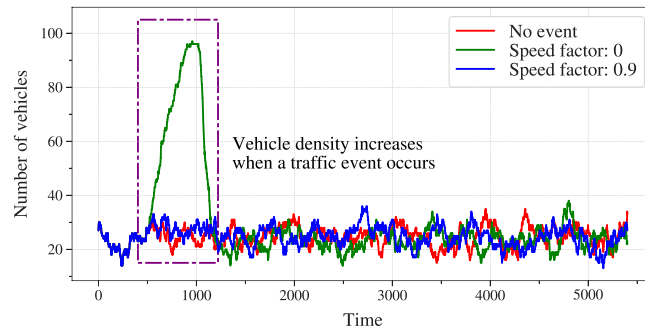


Fig. 9. The vehicle density over time in the vicinity of RSU₁ impacted by a traffic event.

4. **False positive rate:** the ratio between the number of vehicles detected falsely and the number of benign vehicles.

5.2. Detection of false ERM attacks

In this simulation, a traffic event occurs at 500 s and lasts for a duration of 500 s. The coordinate of event is (2000, 1500), which is roughly at the center of map. To simulate the varied impacts of traffic events on the movements of vehicles, each traffic event is associated with a speed factor. Vehicles impacted by the event reduce their speeds using the speed factor as a multiplier. For instance, if an event has a speed factor of 0.5, impacted vehicles will slow down to half of their original speeds. In the simulations, two cases are considered: (i) the speed factor of event is 0, simulating a congestion scenario, and (ii) the speed factor of event is 0.9, indicating that the event has little impact on traffic flows.

Fig. 9 illustrates the different traffic flows in the above two cases. It can be observed that when the speed factor is 0.9, the number of vehicles in the vicinity of RSU₁ impacted by a traffic event is mostly similar to that under normal situations (i.e., no event occurs). In contrast, when the speed factor is 0, the vehicle density rises sharply starting at 500 s. Furthermore, as shown in Fig. 10, an event with a low speed factor renders a high (i.e., close to 1) anomaly index, which indicates that the traffic operation deviates significantly from normal circumstances under the impact of this event. Therefore, if the anomaly index is higher than the threshold η , it is reasonable to determine that an event has occurred.

Our proposed scheme is simulated in both cases and compared with the following schemes:

1. **ML** (Gyawali et al., 2020): In this scheme, a machine learning model is deployed on each vehicle to predict whether an event has occurred. Specifically, the Random Forest is chosen as the prediction model, and traffic parameters such as average speed are utilized as training features. With the prediction results, vehicles detect their neighboring vehicles that report a contradictory event status as malicious. Then, the Dempster–Shafer theory is employed to aggregate detection results from various vehicles.
2. **Towards** (Lv et al., 2022): This scheme combines spatio-temporal information and social relations to distributedly detect malicious vehicles. The spatio-temporal information measures the distance of an ERM to the actual event in terms of time and space, while the social relations are constructed based on the movement of vehicles.

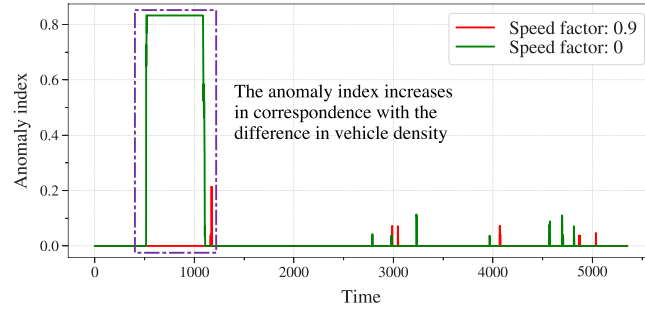


Fig. 10. The anomaly index over time in the vicinity of RSU₁ impacted by a traffic event.

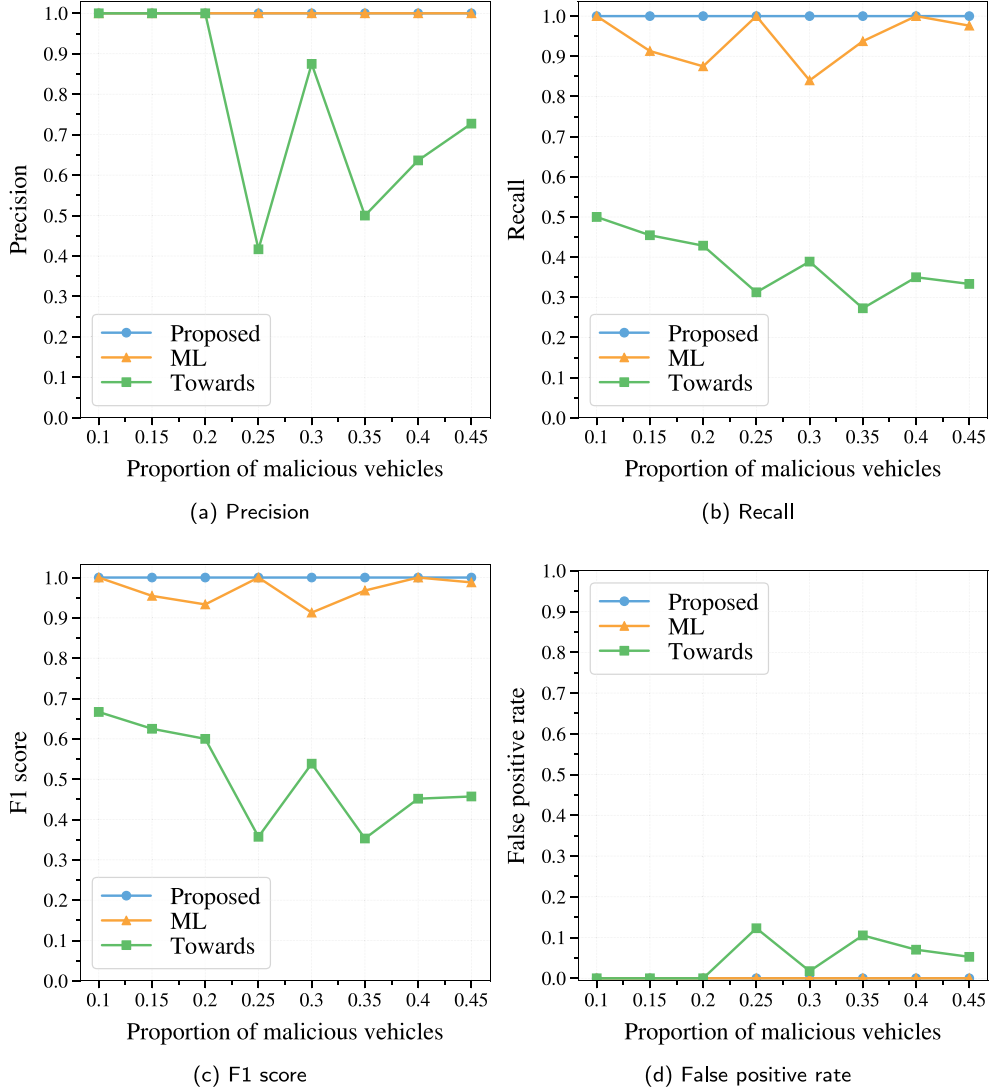


Fig. 11. Detection results when the anomaly index is higher than the threshold.

5.2.1. Case 1: high anomaly index (speed factor 0)

Fig. 11 illustrates the detection metrics under a traffic event with the speed factor 0. In this case, the anomaly index is higher than the threshold, which means the vehicle consensus mechanism is not triggered. Firstly, as shown in Fig. 11(a), both our proposed

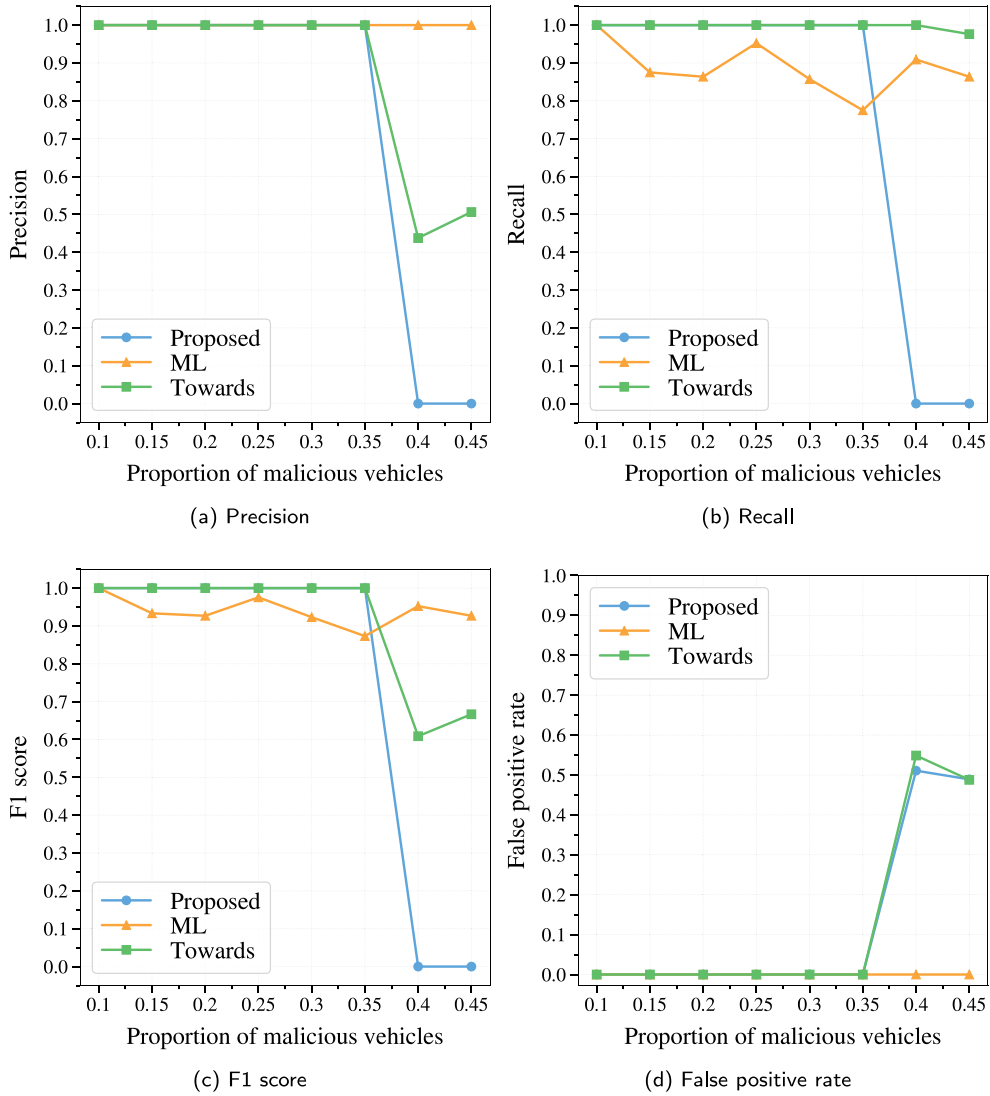


Fig. 12. Detection results when the anomaly index is lower than the threshold.

scheme and ML maintain a 100% precision with all proportions of malicious vehicles. However, the precision of Towards drops significantly when the proportion of malicious vehicles reaches 25%. Secondly, Fig. 11(b) depicts the recall of three schemes. Our proposed scheme achieves a steady 100% recall in all circumstances, while the recall of ML is slightly lower with an average of 94.27%. In contrast, the recall of Towards declines from 50% to around 30% when the proportion of malicious vehicles increases. Thirdly, Fig. 11(d) depicts the results of false positive rate. Only the Towards scheme shows an average false positive rate of 7.37%, when the proportion of malicious vehicles is higher than 25%.

The reason for the performance of our proposed scheme is the accurate prediction of traffic event through the GBDT-based traffic anomaly detections. Although the ML scheme employs Random Forest to predict traffic events, the predictions are made distributedly and not all vehicles result in correct predictions. In simulations where the anomaly index is high, the Random Forest achieves a recall of 98% for the traffic event, rendering a lower recall of the ML scheme. In contrast, the Towards scheme relies on messages received from neighboring vehicles to determine the status of traffic event. In the case with a high anomaly index (i.e., the speeds of vehicles are low), vehicles encounter other vehicles sparsely, thus becoming highly likely to be affected by false event reports. Therefore, the Towards scheme results in a relatively low recall and high false positive rate.

5.2.2. Case 2: low anomaly index (speed factor 0.9)

Fig. 12 illustrates the detection metrics under a traffic event with the speed factor 0.9. In this case, the anomaly index is lower than the threshold, and detections are accomplished by the vehicle consensus mechanism. Firstly, as shown in Fig. 12(a) and Fig. 12(b), our proposed scheme maintains 100% precision and recall when the proportion of malicious vehicles does not

Table 3
Time delay caused by consensus mechanism with different numbers of consensus vehicles.

Number of consensus vehicles	Steps	Delay (each step represents 0.1 s)
3	10	1 s
4	11	1.1 s
5	17	1.7 s
6	20	2 s
7	23	2.3 s
8	25	2.5 s
9	30	3.0 s
10	32	3.2 s
15	52	5.2 s
20	69	6.9 s
25	93	9.3 s

Table 4
Comparison of different anomaly detection models.

Model	MSE	MAE	R2
GBDT	0.458	0.448	0.968
SVM	0.465	0.439	0.968
KNN	1.412	0.921	0.901
DT	0.947	0.659	0.934
LR	0.384	0.360	0.973

exceed 35%. With more malicious vehicles, our proposed scheme fails to correctly identify malicious and benign vehicles, leading to a relatively high false positive rate. Secondly, while the precision of ML is high, the recall is lower than both our proposed scheme and the Towards scheme, achieving an average of 88.70%. Thirdly, the Towards scheme maintains a high recall of 100% throughout. Nevertheless, the precision of Towards drops significantly when the proportion of malicious vehicles reaches 40%, and the false positive rate increases to around 50% at the same time.

In this case, the performance of our proposed scheme conforms to the characteristics of PBFT-based consensus mechanism. When the proportion of vehicles sending false ERMs reaches 40%, the consensus mechanism results in an opinion contradictory to the actual status. In the ML scheme, the Random Forest achieves a recall of 53% in this case, which is lower than that in Case 1 due to the less impact of traffic event on vehicle movements. Correspondingly, the recall of ML is also lowered. Moreover, compared to our proposed scheme, the need to retrain the learning model with malicious data in different cases is the main drawback of the ML scheme. Additionally, the recall of Towards is higher than that in Case 1, since message exchanges between vehicles are more frequent when the speed factor of traffic event is higher. However, when malicious vehicles account for more than 40%, vehicles tend to be misled by false ERMs from their neighbors, rendering a higher false positive rate.

5.2.3. Delay of consensus mechanism

Considering the potential time delay of reaching consensus in traffic event validation, the consensus mechanism is triggered only in situations where the traffic anomaly is insignificant. In the PBFT algorithm, the delay is related to the number of replicas. In this simulation, the delay is measured under 25% malicious vehicles, with a varying number of vehicles in the consensus group. It is worth noting that since the simulation is done with a discrete event simulator, the delay is calculated by the discrete steps needed during simulation (Zhao et al., 2022). As shown in Table 3, the delay increases as the number of consensus vehicles increases. Assuming that each simulation step is in accordance with the broadcast frequency (e.g., 0.1s as specified in IEEE 802.11p), if the number of consensus vehicles is under 15, the consensus mechanism introduces a delay of around 5s. When the number of consensus vehicles reaches 25, the consensus mechanism takes over 9s, which is unacceptable for certain traffic events.

5.2.4. Comparison of anomaly detection models

With the same training data, the GBDT based model is compared with other well-accepted regression algorithms, including Support Vector Machine (SVM), K Nearest Neighbors (KNN), Decision Tree (DT), and Linear Regression (LR). Although experimental results in Table 4 show that LR yields superior results in terms of common regression metrics such as Mean Square Error (MSE), Mean Absolute Error (MAE) and R2 in the training data, in the simulation scenario, LR fails to identify the traffic anomaly through the anomaly index. As shown in Fig. 13, the anomaly index calculated based on LR is consistently lower than the threshold. Therefore, the detection scheme cannot identify false ERMs and malicious vehicles properly. Moreover, the anomaly index calculated based on SVM and KNN incorrectly indicates that a traffic event happens between 4500 s and 5000 s, which will lead to false positives of the detection scheme. Additionally, compared with DT, the GBDT based model yields more accurate prediction results and anomaly index.

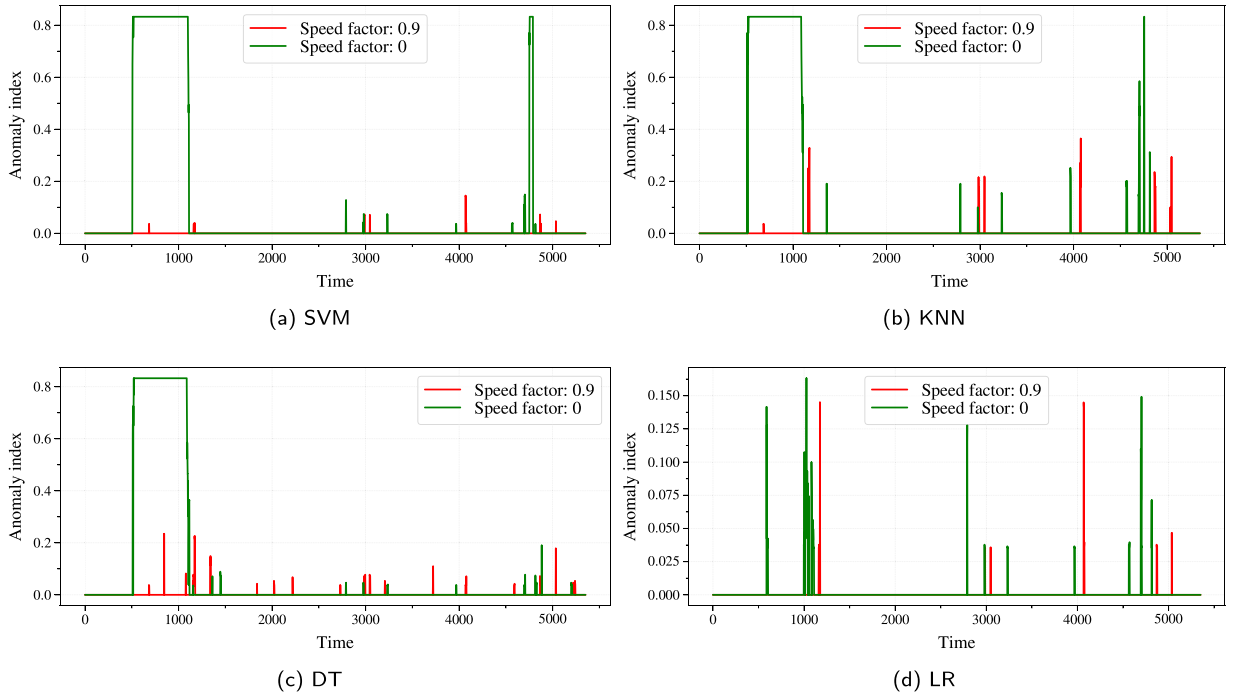


Fig. 13. Impact of different anomaly detection models on false ERM detection.

5.3. Detection of false BSM attacks

In this simulation, malicious vehicles continuously send BSMs with falsified data. Specifically, in false BSMs, the speeds are halved and an offset of 400 m is added to the coordinates. Our proposed scheme is simulated and compared with the following schemes:

1. **Host-based** (Zaidi et al., 2016): In our simulations, this scheme based on hypothesis test is evaluated in both distributed and centralized variations. In this distributed scheme, each vehicle forms its own detections, and the union of detections from all vehicles is taken as the final result. Moreover, considering the timeliness of statistics, detections made before simulation time 5240s are discarded to minimize the false positive rate. In addition, a centralized variation of this scheme is also evaluated, with a central server aggregating and accumulating detections from each vehicle during the simulation.
2. **Trajectory** (Huang et al., 2021): In this scheme, three features in each BSM are discretized and represented as three letters. Thus, each trajectory (consisting of multiple BSMs) is transformed into a sentence with three-letter words. Then, the Word2Vec (Mikolov, Chen, Corrado, & Dean, 2013) algorithm is employed to generate the vector representation of each word. Finally, the falsified and normal trajectories are identified by hierarchical clustering. Specially, the three features utilized in our simulations are speed, longitude, and latitude.
3. **RSO-FDS** (Anyanwu, Nwakanma, Lee, & Kim, 2023): The RSO-FDS mainly leverages the Random Forest model to reliably detect normal and false data. To avoid overfitting of the Random Forest, the Randomized Search Optimization (RSO) algorithm is utilized to obtain optimal model parameters. The Random Forest model is trained using BSMs collected with 50% malicious vehicles, and the trained model is loaded during simulation.

Fig. 14 shows the metrics of false BSM detections. In Figs. 14(a) and 14(b), both our proposed scheme and Trajectory exhibit high precision and recall with up to 40% malicious vehicles. When the proportion of malicious vehicles reaches 45%, the precision and recall of our proposed scheme drops to 96% and 71.64%, respectively. The reason is that these two schemes are based on clustering algorithms, which rely on the assumption that malicious vehicles are fewer than benign vehicles. Therefore, when the proportion of malicious vehicles is close to 50%, the performance of both our proposed scheme and Trajectory decreases. However, the Trajectory scheme exhibits high false positive rates with more than 40% malicious vehicles, which is inferior to our proposed scheme. Since the cluster quality measurement is integrated with the reputation mechanism in our scheme, the false positives are well controlled.

Moreover, the precision of Host-based scheme increases as the proportion of malicious vehicles rises, due to its lower recall and higher false positive rate. In contrast, the centralized variation of Host-based scheme achieves a significantly higher precision when the proportion of malicious vehicles does not exceed 30%. This is because false detections are overwhelmed by correct detections

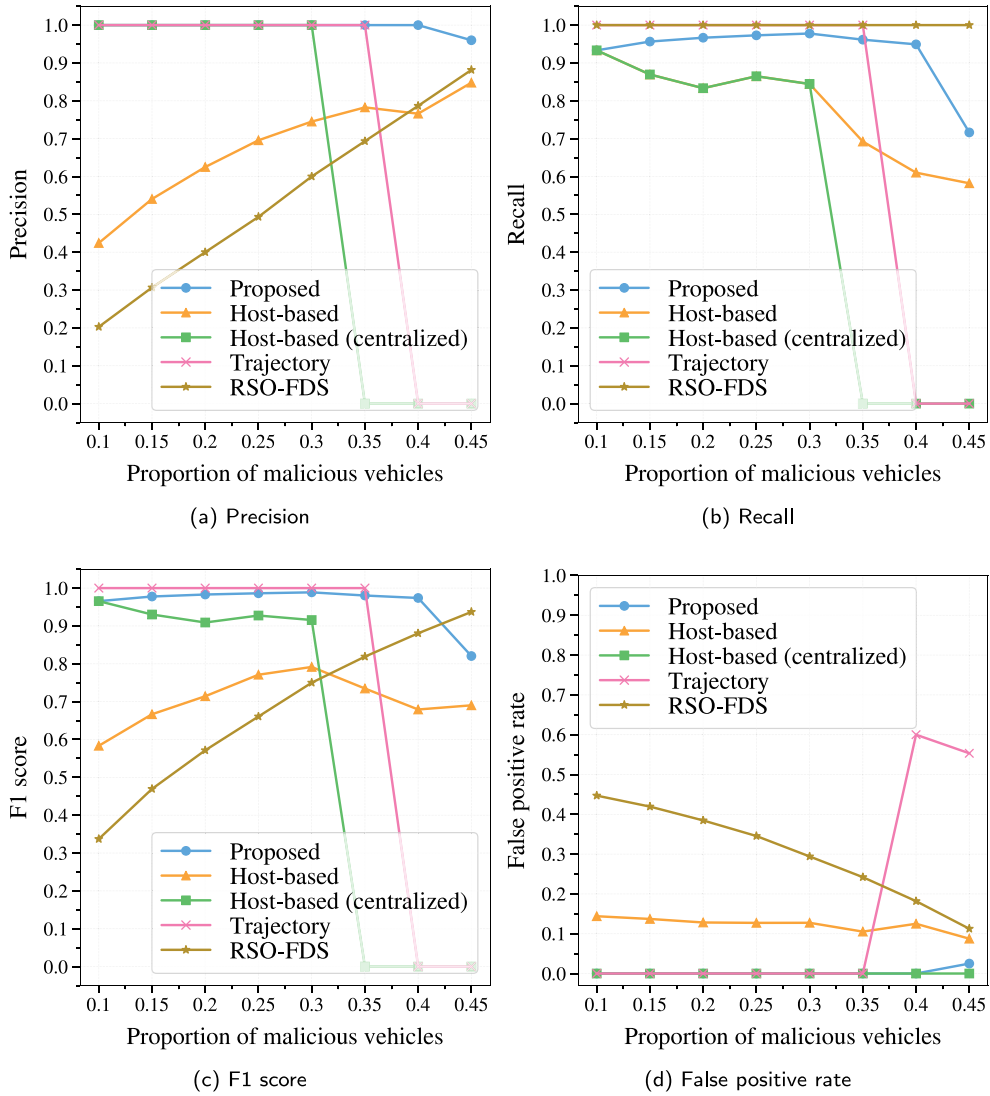


Fig. 14. Detection results of false BSM attacks.

afterward during the whole simulation. However, when the proportion of malicious vehicles increases to 35%, the centralized Host-based scheme fails to malicious and benign vehicles. The reason is that the test statistic used in this scheme is affected by false data, and thus the t-test cannot function as expected.

The RSO-FDS exhibits higher false positive rates and lower precision than other compared schemes. Although the learning model performs well under the situation when there are 50% malicious vehicles, its classification performance decreases when the proportion of malicious vehicles is lower than 50%. The model mistakes vehicles under certain movement states as malicious, yielding a relatively high false positive rates. Therefore, the model has to be retrained to adapt to different proportions of malicious vehicles, which is a major inconvenience in its application compared to our proposed scheme based on clustering. Additionally, the RSO-FDS does not include a vehicle-targeted reputation mechanism that aggregates multiple detection results. Vehicles are considered malicious once their messages are classified as false message, also rendering higher false positive rates.

Fig. 15 illustrates the perceived average speed in the vicinity of RSU_1 with 35% malicious vehicles, with the malicious multiplier applied to real speed values set to 0.1. It can be observed that without the proposed detection scheme, the perceived values are constantly lower than the real ones. However, with the detection scheme filtering out the identified false data, the perceived values do not deviate significantly from real values throughout the simulation. Nevertheless, since the recall of our proposed scheme is below 1 with 35% malicious vehicles, the perceived values with detection still deviates slightly from real values and perceived values with all attackers filtered. The ablation study is conducted for false BSM detection. The cluster quality computing is removed and evaluated against the original scheme under 15%–45% malicious vehicles. Experimental results in Table 5 show that without the cluster quality computing, the precision and recall of our proposed scheme remain unchanged when the proportion of malicious

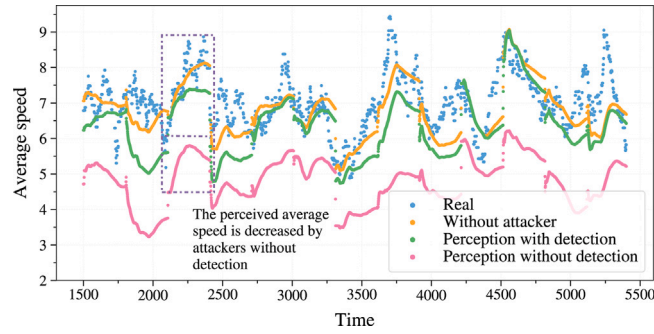


Fig. 15. Perceptions of average speed in the vicinity of RSU_1 with 35% malicious vehicles.

Table 5

Ablation study of false BSM detection with different proportions of malicious vehicles (Precision / Recall / False positive rate).

Proportions of malicious vehicles	With cluster quality computing	Without cluster quality computing
15%	1/0.957/0	1/0.957/0
25%	1/0.973/0	1/0.973/0
35%	1/0.962/0	1/0.923/0
45%	0.96/0.716/0.025	0.932/0.612/0.038

vehicles is under 35%. However, under challenging conditions where the proportion of malicious vehicles reaches 35%, the precision and recall of our proposed scheme decreases, and the false positive rate rises. The cluster quality computing, combined with the reputation mechanism, eliminates the potential impact of inaccurate clustering under high proportion of malicious vehicles, and improves the overall performance of false BSM detection.

5.4. Discussion

5.4.1. Application of GBDT and PBFT

The proposed scheme employs GBDT and PBFT for false ERM detection. The reason for applying GBDT is three-folded. Firstly, in our scheme, the vehicle density in a certain area is treated as time series data for prediction. Existing research (Cao, Wu, Yang & Yin, 2023) has shown that the performance of GBDT-based models is comparative to deep learning models in several time series prediction tasks. Secondly, GBDT-based models are generally more efficient in the predicting phase than deep learning models, especially when devices like Road Side Units usually have limited computation resources. The integration of GBDT in our traffic anomaly detection system helps reduce potential delays caused by the detection process. Thirdly, in our scenario, the time series to predict is short-term and stable, since most traffic events only last for minutes or hours. This kind of time series is suitable for GBDT models. By leveraging the ensemble of decision trees and boosting techniques, GBDT can effectively capture patterns in the vehicle density data, aiding in the identification of anomalous traffic events that may lead to false ERMs.

The consensus algorithms play a complementary role in false ERM detections when the traffic anomaly is insignificant. PBFT is chosen in the proposed scheme as a widely-used and well-established solution for achieving Byzantine fault tolerance in distributed systems (Sankar, Sindhu, & Sethumadhavan, 2017). Moreover, a reputation-based selection mechanism is proposed to limit the number of vehicles participating in the consensus algorithm, which improves the overall efficiency of our false ERM detection. It is also worth noting that the choice of consensus algorithms is not limited to PBFT. The proposed detection scheme has the adaptability of employing different consensus algorithms. The PBFT is chosen as a well-accepted algorithm to demonstrate the feasibility of our proposed consensus-based detection scheme.

5.4.2. The relation of false BSM and ERM detection

In the IoV, ERMs and BSMs are not necessarily sent concurrently. On the one hand, BSMs are messages sent periodically, which serves as beacons signaling the status of vehicles. On the other hand, ERMs are sent when vehicles perceive the occurrence of certain traffic events with onboard sensors. It is possible that BSMs are sent without corresponding ERMs. Therefore, in the proposed scheme, the detection of false ERMs and BSMs are designed to work separately. Nevertheless, there is indeed one aspect in which BSMs and ERMs may be coupled, which lies in the use of vehicle density data in false ERM detections. There exist multiple sources for the data: (i) the transportation infrastructure, such as sensors and cameras installed at intersections, and (ii) BSMs sent by vehicles. In the latter case, vehicles could send BSMs with falsified identity to create the illusion of higher density, which forms a Sybil attack. However, the detection of Sybil attacks is out of the scope of our paper and there are ongoing research on this specific topic (Ayaida, Messai, Najeh, & Boris Ndjore, 2019).

5.4.3. Multiple overlapping events

In real traffic scenarios, it is possible that there are multiple events happening concurrently with the coverage range overlapping with each other. The proposed scheme is capable of handling this situation for the following reasons. Firstly, our scheme aims at traffic event such as accidents or traffic jams. This kind of events does not occur frequently in a certain area. Existing research on traffic event prediction even needs to design specific techniques to tackle the sparsity of traffic event data (Trirat, Yoon, & Lee, 2023). Therefore, in the rare cases that overlapping events happens at the same time, the number of events should be low. Secondly, our scheme treats each event separately. Vehicles send ERMs for each event identified by the location of event perceived by the onboard sensor. Multiple detection process may be initiated and running concurrently. Thirdly, events can be identified by its location. If an RSU received multiple ERMs reporting events in close proximity (e.g., a pile-up involving multiple vehicles) or with overlapping coverage range, it can group the events into one event with higher priority and initiate the detection process on the event.

5.4.4. Scalability and practical applicability

The proposed scheme is adaptable to large-scale IoV environments, since it mainly relies on RSUs to implement false message detections. Each RSU is responsible for messages sent by vehicles in its covering region. This distributed manner is critical in the situation of large-scale IoV environments. The central server is only utilized for computing the reputation score of vehicle, which is within the capability of modern cloud servers. Moreover, the proposed scheme employs techniques like statistical models, clustering, and machine learning, which are efficient in identifying deviations from normal behavior. In real applications, technologies such as edge computing can be integrated to leverage nearby servers for the training of machine learning models, which improves the scalability of our proposed scheme.

Moreover, in practice, privacy concerns are common among IoV users. While our proposed scheme does not include a privacy mechanism, existing research has explored various technologies for privacy protection in IoV (Chen et al., 2022; Zhang, An et al., 2024), which can be employed in the computation process of IoV applications. Moreover, since our proposed scheme relies on traffic data, mechanisms to tackle potential traffic data loss can also be integrated (Zhang, Zhang, Yan, Qiu & Gao, 2021). Lastly, message dissemination between V2X devices plays an critical role in our proposed scheme, where existing research such as Chen, Mao, Li, and Zhang (2020), Zhang, Cui and Zhang (2019) and Zhang, Wang, Zhang, Zhu et al. (2023) can serve as a technological foundation.

6. Conclusion

Aiming at ensuring the information integrity of IoV, this paper proposes a detection scheme targeting both false ERMs and false BSMs. Considering the varied impact of traffic events on vehicle movements, our proposed scheme integrates GBDT-based traffic anomaly detection with a PBFT-based vehicle consensus mechanism. The anomaly index is proposed based on the divergence between predicted and observed traffic data, quantifying the impact of traffic events. If the impact is significant, the detections are made based on the predicted status of traffic event, minimizing the time delay and communication overhead of consensus mechanisms. Otherwise, the vehicle consensus mechanism is proposed to determine the event status within selected consensus vehicles. Furthermore, for false BSM detections, a cluster quality indicator based on hypothesis tests is proposed to complement the K-means++ clustering algorithm, assisting in the long-term vehicle evaluation through the reputation mechanism. Simulations demonstrate that our proposed scheme exhibits excellent effectiveness in identifying false messages and malicious vehicles in the IoV. For false ERMs, our proposed scheme achieves 100% precision and recall when the traffic event is significant, and when the traffic event is insignificant, the same result can be maintained with less than 35% attackers. For false BSMs, our proposed scheme exhibits over 90% precision and recall with up to 40% malicious vehicles. While the recall drops to around 71% with 45% malicious vehicles, the false positive rate of our scheme is still low (i.e., around 5%). Our future research will focus on more advanced models for traffic anomaly detections and traffic event predictions, which will in turn improve the performance of false message detections. For instance, existing research (Trirat et al., 2023) in the transportation area has demonstrated that additional features such as demographic data, road network, and regional functions can improve the accuracy of traffic prediction. Moreover, microscope driving behaviors, such as sudden braking and lane changing, can also contribute to traffic anomaly detections. These features can be further integrated into the traffic anomaly detection model.

CRedit authorship contribution statement

Yu'ang Zhang: Writing – original draft, Software, Methodology, Investigation, Data curation. **Chaklam Cheong:** Writing – review & editing. **Sifan Li:** Writing – review & editing. **Yue Cao:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Xiaoling Zhang:** Resources. **Dong Liu:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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